

MULTIPLE CHOICE TEST TOPICS

1005 of ready-to-use topic questions!



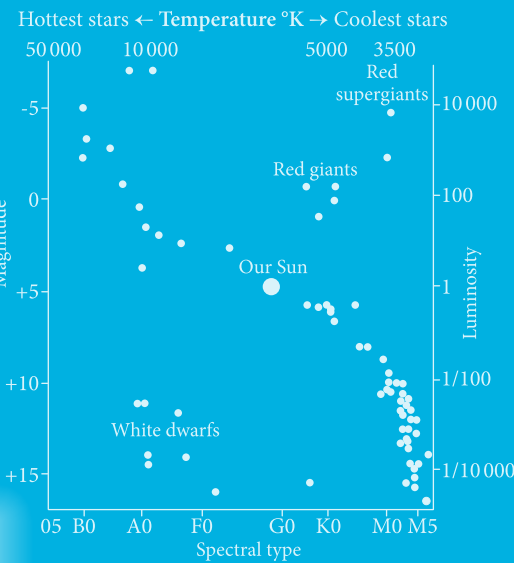
Locality 2



Locality 3



Astronomy Earth and Environmental Science



Zenda Spry-Pohl
Raimund Pohl

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Astronomy, Earth and Environmental Science

**Zenda Spry-Pohl
Raimund Pohl**

This book was commissioned by Jennifer Gough, production was by Piper Print Media, and David Dickson supplied the illustrations.

Zenda Spry-Pohl has more than 35 years of experience teaching science in secondary schools in NSW. She has taught Junior Science, Biology and Geology for many years in Tamworth, Killara and Cheltenham, and in other schools for shorter periods. Zenda has recently completed extra university courses to update her knowledge base. She has been a marker for HSC Geology and has considerable experience in preparing examination material and worksheets and booklets. Zenda has a keen interest in astronomy and has visited radio telescopes in Australia and Nancy, France. She has been a head teacher of welfare and has a particular interest in ensuring that individuals have work at a level appropriate to their needs. Zenda is the co-author, with her husband Raimund, of *A-Z of Science* books 3 and 4 for Blake Education.

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INTRODUCTION

This series of photocopiable test topic books has been written to assist teachers to set tests and exams for students in years 9–10. The pages and questions can also be used:

- ✱ to prepare students for this specific type of assessment in more senior year levels
- ✱ as a diagnostic tool to check students' understanding of a just completed topic
- ✱ for homework, extension and revision.

The questions in this series ask students to: recall or remember a fact, make calculations, interpret data from a table, make sense of a diagram, read off values from a graph, forecast what will happen from a graph and extract information from a written passage.

Each question could be answered in the normal test manner or students could:

- ✱ be asked to justify their response
- ✱ describe why the wrong answers are incorrect
- ✱ provide examples they know that illustrate and support the correct answer.

Teachers should advise students how to record the answers – such as circling the best answer or answering on a separate answer grid. A list of helpful tips for answering multiple choice questions has been provided and was written by teachers well used to setting and marking tests and preparing their students for examinations.

An editable version of all the questions is also available on the CD-ROM that is attached to the inside back cover of this title. All material can be edited only in accordance with the user agreement.

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Before the test or exam

- ✘ Make sure you have planned out how much time you have to answer the multiple choice section of the exam and know exactly how many marks this section is worth.
- ✘ Do as many practice multiple choice questions from past exams as you can.
- ✘ Ensure that you know the meaning of common words found in exams such as *all, many, some, sometimes, always, never, none, not, few*. Also know the definitions of key words such as *describe, analyse* and *evaluate*.

Reading test questions

- ✘ Read through the paper, and select the easiest questions to do first. Do the hardest questions last. This gives you confidence to do your best.
- ✘ Read each question and try to answer it without looking at the choices. Then read through the choices. Doing this allows you to make a clearer and more accurate choice.
- ✘ Be alert for grammatical inconsistencies between the question and the potential answers. A choice is nearly always wrong if the question and the answer don't combine to make a grammatically correct sentence.
- ✘ Don't read anything into or out of the question, or assume it has been written to trick you.
- ✘ Some questions are tricky though. For example, *Which of the following is not ...; All of the following except ...* A response may be true, but not the answer to the question set.
- ✘ The wrong answers are called 'distractors' – there's a reason for this name!

Strategies

- ✘ Eliminate answers you know aren't right and be logical in your approach.
- ✘ With the harder questions, delete the answers that are least likely to be right. Your chances of selecting a correct answer increases with every alternative you eliminate.

- ✘ If there is no penalty for guessing, and you cannot work out the right answer, make learned guesses.
- ✘ Never leave a multiple choice question unanswered.
- ✘ Don't be put off a response just because it seems too obvious or too simple.
- ✘ If two responses appear to be equally correct, eliminate the response that appears to be least related to the question being asked.
- ✘ Use a common sense approach to numerical questions. First think out whether the answer will be in 10s, 100s, 1000s or millions.
- ✘ Unless otherwise instructed, you may write your workings out or reasoning on the test paper and this will not be looked at by the marker.

At the end of the test or exam

- ✘ Leave yourself time at the end of the exam to check your answer sheet to see if you have recorded your answers correctly.
- ✘ Do not change answers unless you know that your recorded answers are incorrect.
- ✘ Do not change your answers just because there are too many or too few of a particular letter on your answer sheet.

After you have been given your marked answer sheet

- ✘ Look carefully at the questions you answered incorrectly. If you answered them wrongly because you did not understand the content, then it is 'back to the books'.
- ✘ If you made an error because your multiple choice test technique was poor, then re-read the tips for answering this kind of question and do more practice.
- ✘ If anxiety meant that you did not perform as well as you wanted, remember that sound planning and a good test technique will help you to feel more confident in any exam. Practice boosts confidence and lessens anxiety.

- 1** Which is the best description of a solar system?
- A** Planets and stars that orbit a large planet.
 - B** Planets and rocks that orbit a single sun.
 - C** Stars and planets that orbit a galaxy.
 - D** Satellites and planets that orbit Earth.
- 2** Why does night occur?
- A** The Moon blocks out the Sun.
 - B** The Earth rotates on its axis as it orbits the Sun.
 - C** Darkness is a natural occurrence.
 - D** The Sun moves under the Earth as it revolves.
- 3** Which interactions cause the seasons of the year?
- A** The distance between the Sun and the Earth as the Earth revolves around the Sun
 - B** The huge difference in distance to the Sun caused by the Earth's tilted axis
 - C** A spread of light over an area as the Earth comes closer to the Sun
 - D** The length of daylight hours as one hemisphere is tilted towards the Sun
- 4** Which events are typical of the spring equinox?
- A** The days are much longer than the nights.
 - B** The weather is cooling.
 - C** Flowers blossom, as the mornings are early.
 - D** The days are the same length as the nights.
- 5** Which statement best describes the Sun?
- A** The major source of energy in the solar system
 - B** A mixture of gases, mainly hydrogen and oxygen
 - C** The centre of our galaxy
 - D** A yellow gaseous nebula
- 6** Mars is similar to the Earth in many ways. What features do they have in common?
- A** They are the same size and have the same gravity.
 - B** Mars has an atmosphere, polar caps and a rocky surface.
 - C** They both have an atmosphere that could support human life.
 - D** Mars has an atmosphere and oceans.
- 7** Life as we know it has possibly only evolved on Earth because:
- A** the atmosphere is composed of only nitrogen and the temperature is mild.
 - B** there are quantities of nitrogen, oxygen and saline water for life.
 - C** Earth's atmosphere is composed of only oxygen and carbon dioxide.
 - D** nitrogen, oxygen, carbon dioxide and moderate temperatures exist.
- 8** What causes the phases of the Moon?
- A** The Earth rising.
 - B** The Moon's position in relation to the Earth as it orbits the Earth.
 - C** The Moon's position in relation to the Sun as it rotates on its axis.
 - D** Eclipses caused by reflected light.

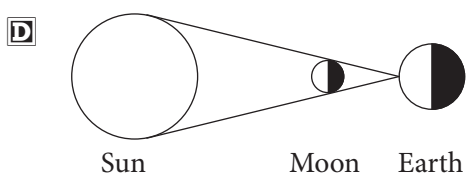
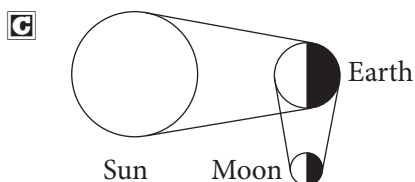
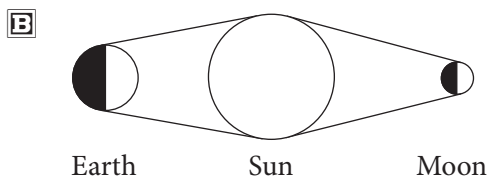
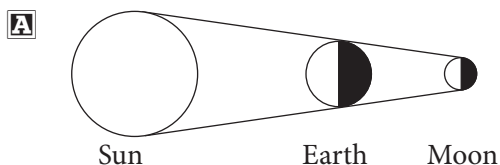
9 The 'seas' on the Moon are caused by:

- ☐ A solid, dry ice accumulations.
- ☐ B volcanic eruptions close to the surface.
- ☐ C asteroid bombardment.
- ☐ D ancient lava outpouring from the interior of the Moon.

10 The Moon and the Earth have mass. What prevents the Moon from colliding with the Earth?

- ☐ A Gravitational pull
- ☐ B Rotational speed
- ☐ C Gravitational pull and orbital velocity
- ☐ D Gravitational pull and rotational speed

11 Which diagram best describes a solar eclipse?



12 What causes the tides on the Earth?

- ☐ A The Earth rising in space
- ☐ B The gravitational pull of the planets
- ☐ C The Sun's and Moon's positions in relation to the Earth
- ☐ D Solely the gravitational pull of the Moon on the water

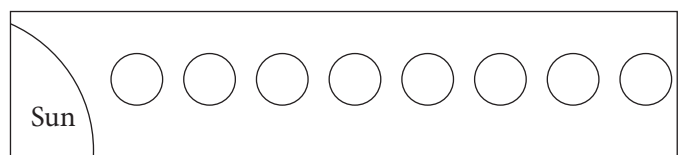
13 Which statement best describes asteroids?

- ☐ A Large particles of ice that orbit the Sun between Earth and Mars
- ☐ B Small particles of ice and rock that orbit the Earth
- ☐ C Large rocks that orbit the Sun between Mars and Jupiter
- ☐ D Small rocks that orbit the Earth

14 Which is the best description of comets?

- ☐ A A central mass of rock and a misty envelope that orbits the Earth
- ☐ B Large particles of ice or rock that have elongated orbits around the Sun
- ☐ C Large particles of ice or rock that orbit the Earth
- ☐ D The result of human-made space junk that orbits the Earth in an ellipse

15 The following diagram represents the planets and the Sun.



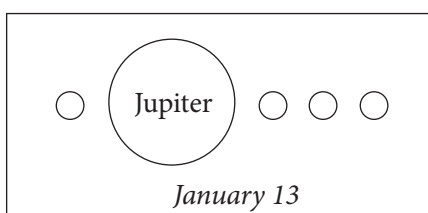
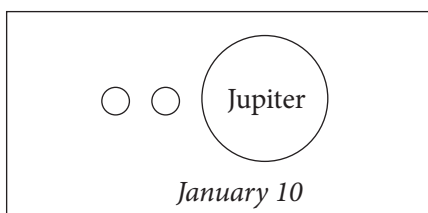
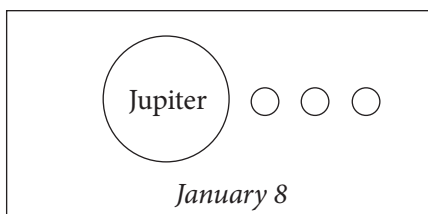
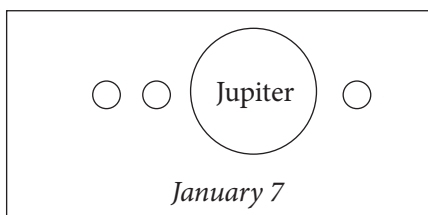
Which feature is reasonably represented in the diagram?

- ☐ A The size of the planets relative to each other
- ☐ B How many planets exist in the solar system
- ☐ C Accurate distances between the planets
- ☐ D The scale of the planets relative to the Sun

16 To represent the distance between planets accurately, what information would you need to know?

- ☐ A How far it is between planets, in kilometres
- ☐ B The definition of a light year
- ☐ C The definition of an Astronomical Unit
- ☐ D How wide your page is, in centimetres

17 In 1610, Galileo observed four objects that revolved around Jupiter. He called these objects moons.



From his observations, he concluded that:

- ☐ A all planets have moons.
- ☐ B bigger planets have more moons.
- ☐ C Ptolemy's geocentric model of the solar system was physically possible.
- ☐ D Copernicus' heliocentric model of the solar system was physically possible.

18 Which idea was proved *wrong* by Galileo's observations of Jupiter?

- ☐ A All objects in space orbit Earth.
- ☐ B Many objects in space have circular orbits.
- ☐ C Planets orbit the Sun.
- ☐ D Jupiter has several moons.

19 Earth 'laps' those planets that have a slower orbit speed. What effect will an observer on Earth see in the sky?

- ☐ A Planets appear to reverse direction.
- ☐ B Slower planets will always be behind the Earth.
- ☐ C Slower planets will appear much dimmer.
- ☐ D Planets will suddenly appear on the other side of the Sun.

20 If you were creating a general definition of a planet, what criteria would be best astronomically?

- ☐ A A planet must orbit a star and be massive enough for its own gravity to form a spherical shape.
- ☐ B A planet must be made of rock and be bigger than Neptune.
- ☐ C A planet must orbit the Earth, be bigger than Neptune and have moons.
- ☐ D A planet must orbit the Sun annually and have enough substance to form a spherical shape by its own gravity.

- 1** Our solar system lies in our galaxy, the Milky Way. There may be 200–400 billion other main objects in the galaxy. What are these objects most likely to be?
- ☐ A Planets
 - ☐ B Moons
 - ☐ C Galaxies
 - ☐ D Stars
- 2** A constellation is best described as a:
- ☐ A set of single stars.
 - ☐ B formation of stars in no fixed pattern.
 - ☐ C group of stars in a fixed pattern.
 - ☐ D single set of suns.
- 3** Which of the following is *not* a constellation?
- ☐ A The southern triangle
 - ☐ B Pisces
 - ☐ C Sirius
 - ☐ D Orion
- 4** Our galaxy is called the Milky Way. Why was it given this name and when?
- ☐ A It looks like a diffuse meteor shower and was named by Galileo.
 - ☐ B It was seen from above by a space probe and named in the 1970s.
 - ☐ C It was seen as a sacrifice to the gods and was named by the Aztecs.
 - ☐ D It appears as spilt milk in the night sky and was named in ancient times.
- 5** Galileo looked at distant areas of the Milky Way with his telescope in the 1600s and realised that:
- ☐ A it was made up of thousands of individual stars.
 - ☐ B it is a separate galaxy.
 - ☐ C it is one of a great number of galaxies in the universe.
 - ☐ D the planets move among the stars.
- 6** Which is the best description of the shape of our galaxy?
- ☐ A A sphere with a dense centre
 - ☐ B A flattened Catherine wheel disc with a bulbous centre
 - ☐ C A football shape with onion-like layers
 - ☐ D An irregular grouping with rays into space
- 7** When you see the Milky Way, what are you actually looking at?
- ☐ A The universe of billions of stars
 - ☐ B The solar system and the asteroid belt
 - ☐ C Gas clouds and nebulae
 - ☐ D A view of the plane of our own galaxy
- 8** Which is the best description of the way the stars in our galaxy move?
- ☐ A They spread outwards like fireworks.
 - ☐ B They mill around like hot liquid in a saucepan.
 - ☐ C They orbit around the centre, similar to the way planets orbit around the Sun.
 - ☐ D They move in the same way convection currents move in the ocean.

9 You can see M31, the Andromeda Galaxy, with the unaided eye and more clearly with an amateur telescope. What is M31?

- A** A separate galaxy outside the Milky Way
- B** A part of a separate universe
- C** A cluster of stars in our galaxy, which was given this special name
- D** A group of planetary nebulae named in honour of a Greek goddess

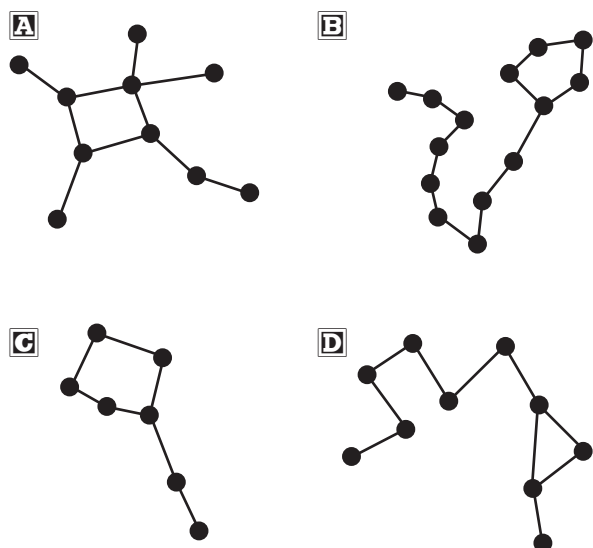
10 A galaxy is best described as:

- A** a group or cluster of stars that moves in space.
- B** a group or cluster of stars that makes a pattern.
- C** a swirling mass of gravity.
- D** a swirling cluster of planets.

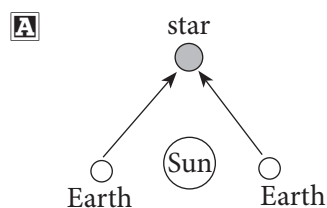
11 A light year is a measurement based on the:

- A** speed of light hitting a planet.
- B** distance that light travels in one year.
- C** time it takes sunlight to reach the Earth.
- D** speed of reflected sunlight.

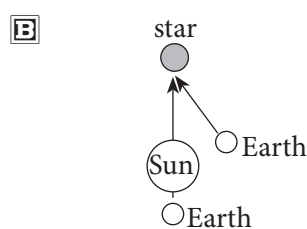
12 Which of these star patterns is used to indicate due south?



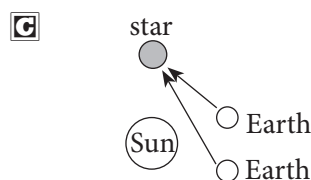
13 Parallax measurements are ways of measuring the distance of stars from the Earth. The model requires that the measurements be made equally to measure the variation between the two sightings. Which of the following would be the best way to use the parallax model?



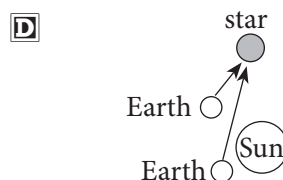
Earth, six months apart



Earth, four months apart



Earth, two months apart



Earth, three months apart

14 The distribution of hydrogen clouds has revealed that the Milky Way is a spiral galaxy of Hubble type Sb or Sc. This has been discovered by:

- A** radio astronomical investigations.
- B** visual observations from spacecraft.
- C** photographic evidence from space.
- D** reception of digital images from a wide array of ground-based telescopes.

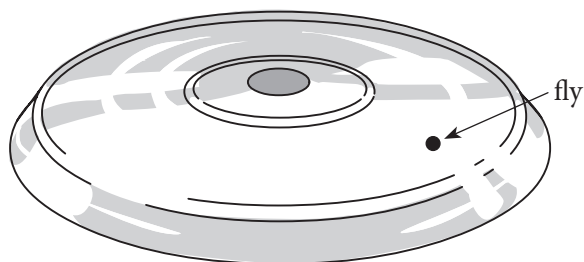
- 15** If there is other life in space, where is it likely to exist?
- A** On Mars
 - B** On nearby stars similar to our own
 - C** On planets similar to our own, orbiting stars
 - D** On stars in far and distant galaxies

- 16** We explore space at great cost and risk. Which reason for exploration is *not* true?
- A** We need to leave Earth in the next 50 years before an asteroid hits us.
 - B** Humans have an innate desire to explore.
 - C** We may find new ways of using our information.
 - D** We can develop new materials and methods of transport.

- 17** As a galaxy, the Milky Way probably has a mass between 750 billion and one trillion solar masses, and its diameter is about 100 000 light years. How would it be described, compared to other galaxies?
- A** One half of a small, paired galaxy
 - B** A giant galaxy in space
 - C** An average galaxy in space
 - D** An elliptic section of the Andromeda galaxy

- 18** The Large and Small Magellanic Clouds, at 179 000 and 210 000 light years respectively, are positioned:
- A** at the far reaches of the Universe.
 - B** just outside the Milky Way.
 - C** in the dust clouds of the Milky Way.
 - D** near the edge of the solar system.

- 19** A good analogy to explain how we see the Milky Way is represented by a fly inside a Frisbee, two-thirds of the way out from the centre.



If you looked outwards from the fly's position, you would see a small amount of space. If you looked inwards, you would see huge amounts of space. What would you see if you looked up or down?

- A** The whole length of the structure
 - B** The bulge in the centre
 - C** A huge amount of space above you
 - D** Only a small amount of space above you
- 20** The Milky Way is our home in the Universe. Practically everything we see in the night sky with the unaided eye is a part of the Milky Way, except for:
- A** almost all the objects in the Messier catalogue.
 - B** globular clusters, 10 000 light years away.
 - C** the Virgo cluster, 50 million light years away.
 - D** planetary nebulae, 1000 light years away.
- 21** There are areas in our galaxy that appear dark and empty in visible light, but with infra-red light we can see:
- A** the disintegration of galaxies.
 - B** the disintegration of meteorite belts.
 - C** black holes forming.
 - D** new stars forming.

1 One technological advance that has resulted from space programs is:

- ☐ A the convection oven.
- ☐ B the Velcro strip.
- ☐ C fibreglass.
- ☐ D nylon.

2 The first person in space was:

- ☐ A Neil Armstrong.
- ☐ B John Glenn.
- ☐ C Harrison Schmidt.
- ☐ D Yuri Gagarin.

3 Weightlessness in space is caused by:

- ☐ A falling mass.
- ☐ B reducing the amount of matter in an object.
- ☐ C lack of gravity.
- ☐ D reducing density.

4 Satellites fall to Earth because:

- ☐ A they orbit at right angles.
- ☐ B they travel sideways, following the Earth's orbit.
- ☐ C the gravity of the Moon pushes them down.
- ☐ D there is a loss of forward speed.

5 The Spitzer Space Telescope is capable of studying the universe at near- to far-infrared wavelengths and gathers information from distant space. It is most likely to be located as a:

- ☐ A satellite in an Earth-trailing solar orbit.
- ☐ B satellite in a geostationary orbit.
- ☐ C ground-based radio telescope.
- ☐ D ground-based visual observatory.

6 There are five general types of satellites with special properties. Satellites for planetary or astronomical studies:

- ☐ A would have orbits that are lower than the orbits of geostationary satellites.
- ☐ B would be global positioning satellites in high orbits.
- ☐ C would be positioned at fixed points above the Equator.
- ☐ D would be geostationary satellites that point away from the Earth.

7 Satellites were sent into space rarely in the 1950s and 60s. Now they are common and are sent up by many countries. Which is the most likely number of satellites that are operational and past their useful life, orbiting Earth?

- ☐ A 200
- ☐ B 1 000
- ☐ C 12 000
- ☐ D 100 000

Questions 8 and 9 refer to the table below.

Table of artificial satellites

Year orbited	Name	Accomplishments
1957	Sputnik 1	First artificial satellite
1958	Explorer 1	First American satellite; discovered Earth's radiation belts
1962	Telstar 1	Relayed first TV pictures across the Atlantic Ocean
1962	Ariel 1	First international satellite (American and British)
1978	Navstar 1	First satellite of future Global Positioning System
1975	GOES 1	First geosynchronous weather satellite
1979	MAGSAT	Mapped the magnetic field near Earth
1990	Pegsat	First satellite launched from an aeroplane in flight
1990	SPOT 2	Maps Earth's resources at high resolution
2003	ICESat	Measures the thickness of Earth's polar ice
2005	STPR1Streak	A technology demonstrator for low Earth orbit satellites
2007		Too many to name

- 8** The early satellite launches were mostly concerned with:
- A** successful satellite launching.
 - B** aiding Global Positioning Systems.
 - C** gaining intelligence about other countries.
 - D** sending instantaneous sports coverage.

- 9** Recent satellites have enabled us to have:
- A** magnetic fields mapped.
 - B** people on the moon.
 - C** Global Positioning Systems.
 - D** inter-country co-operation.

- 10** Most satellites are geostationary. This means that they are:
- A** moving at the speed of sound above the Equator.
 - B** moving in a small orbit above the Poles.
 - C** staying still and not moving around the Earth.
 - D** moving at the same speed as the Earth's rotation, above the Equator.

- 11** As geostationary satellites require exact orbits, their locations may differ only by:
- A** latitude away from the Equator.
 - B** longitude around the Earth.
 - C** distance above the Earth.
 - D** distance from the Poles.

- 12** The International Space Station (ISS) was first crewed in 2000, and since then 17 space shuttles have delivered different astronauts to the ISS. Why do the astronauts only stay for 3–6 months?
- A** Not enough oxygen.
 - B** Too much radiation.
 - C** Not enough experiments to complete.
 - D** Loneliness and poor food.

- 13** When you watch live news reports on television from Europe, you will see a time lag between the interviewer in Australia asking the question and the person in Europe answering. This is due to:
- A** the electromagnetic waves taking a quarter of a second to go to and from a satellite.
 - B** the sound waves reflecting from the ionosphere.
 - C** the electromagnetic waves from the microphone travelling through the cables.
 - D** the sound waves from the TV station taking a quarter of a second to go to and from a satellite.

- 14** Australian observatories at Parkes, Coonabarabran, La Trobe and Woomera provide valuable information for the world's astronomers. This is because:
- A** we have the biggest telescopes in the world.
 - B** our telescopes can cover a larger amount of the skies than the ones in Europe or Asia.
 - C** there are not many observatories in the Southern Hemisphere in our area.
 - D** our telescopes are much newer than the ones in Russia or America and can 'see' more clearly.

- 15** There have been hundreds of space probes sent out to planets in the last 40 years. Which is a likely outcome of the information received?
- A** We can find out if there is life on other planets.
 - B** More accurate geology of the planets will be established.
 - C** We will be able to establish bases on the planets in the next 10 years.
 - D** The question about whether Pluto is a planet or a small rock can be answered.

16 In 1970, the Venera 7 Probe (USSR) made the first successful landing on another planet. Which planet was most likely to be the target?

- ☐ A Mercury
- ☐ B Venus
- ☐ C Sun
- ☐ D Pluto

17 The Galileo Probe to Jupiter was launched jointly by America and Germany in 1995. It was the first probe to enter Jupiter's atmosphere. Which of the following was most likely its task?

- ☐ A To validate Galileo's hypothesis about Jupiter having four moons.
- ☐ B To count the number of moons of Jupiter.
- ☐ C To determine the chemical composition of the Jovian atmosphere.
- ☐ D To collect and return samples of Jupiter to Earth.

Questions 18 and 19 use the following information.

As a spacecraft flies by a planet, its velocity can be increased to deliver it more quickly to the next planet. Voyager's flight to Neptune was reduced from 30 years to 12 years by this manoeuvre.

18 Another term for the manoeuvre described above is:

- ☐ A passing over.
- ☐ B gravitational slingshot.
- ☐ C gravitational resist.
- ☐ D passing velocity.

19 The main advantage of this assisted technique is to:

- ☐ A save fuel, time and expense.
- ☐ B move closer in to the planet to gain better pictures.
- ☐ C avoid moons and asteroids in space.
- ☐ D absorb energy from the planet as the spacecraft moves closer.

20 In 2008, the great science fiction writer Arthur C. Clarke died. He predicted the use of satellites and space travel. Which of the following is a reasonable generalisation about science fiction?

- ☐ A Science fiction is fiction and has no basis in reality.
- ☐ B People often imagine a concept and then turn it into a project for research.
- ☐ C People invent objects as they work through other projects, and then find uses for them.
- ☐ D All science fiction predictions will come true.

21 NASA's two Voyager spacecraft were launched in 1977 to tour the outer solar system. They are now far beyond the orbits of the outermost planets and heading towards interstellar space. Which objects in the solar system have they passed so far?

- ☐ A Jupiter, Saturn, Uranus, Neptune and the solar system boundary
- ☐ B Mars and Jupiter, and they are heading towards Pluto
- ☐ C Uranus, Neptune, Pluto and the nearest star Beta Centauri
- ☐ D Jupiter and all its moons

1 Ancient cultures each believed that the universe was created in a specific way. Which culture believed that the Rainbow Serpent created the universe?

- ☐ A Egyptians near the Nile River
- ☐ B Greeks and Romans 3 000 years ago
- ☐ C Aborigines of Australia
- ☐ D Hindus from India

Questions 2 and 3 are based on the following information.

The universe is made of atoms that have nuclear binding energy. This energy can be released in two ways to provide vast amounts of energy.

2 What is one way this energy can be released?

- ☐ A Fusion from the combination of lighter elements into heavier ones
- ☐ B Combining uranium atoms
- ☐ C Fusion from the combination of heavier elements into lighter ones
- ☐ D Combining oxygen and magnesium to give white light

3 What is the other way that large amounts of energy can be released from atoms?

- ☐ A Division of molecules of planetary material into smaller ones
- ☐ B Fission from the splitting of lighter elements into heavier ones
- ☐ C Fission from the splitting of heavier elements into lighter ones
- ☐ D Division of molecules of the compound water by electrolysis

4 Knowledge of the structure of the atom is one of the basic pieces of information we need before we can understand the universe. Which is the best description of an atom?

- ☐ A Electrons orbiting a nucleus of protons and neutrons
- ☐ B Electrons moving at the speed of light from atom to atom
- ☐ C Protons, neutrons and electrons in a tight ball
- ☐ D Protons, neutrons and electrons orbiting around each other

5 The big bang theory is:

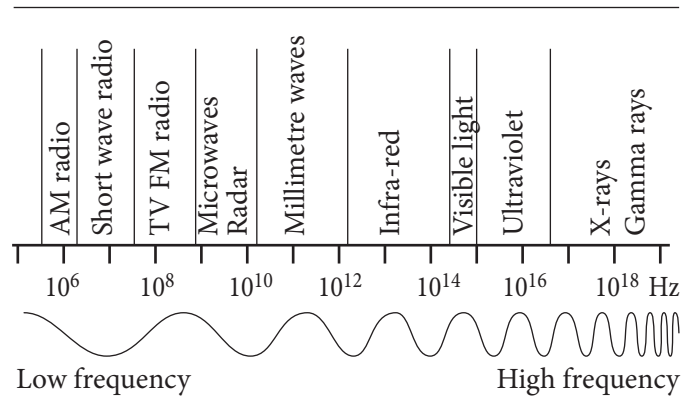
- ☐ A a model that demonstrates a steady state universe.
- ☐ B a model that demonstrates that the universe oscillates.
- ☐ C a theory reinforced by images of the earliest moments, which have recently been recorded.
- ☐ D a theory to explain the beginning of the universe.

6 The big bang theory suggests that a very hot, dense, condensed form of matter exploded many billions of years ago and that this matter has since travelled rapidly outwards. What remnant echo has this left throughout space?

- ☐ A The universe of objects
- ☐ B The stars with their hydrogen
- ☐ C A galaxy in the shape of a pincushion
- ☐ D The cosmic microwave background

- 7** Imagine a balloon with marker pen dots on it being inflated. The dots would spread out as the balloon inflates. This is used as an analogy for how the universe expands. In 1929, the American astronomer Edwin Hubble discovered that most of the galaxies in our universe are moving away from each other. When do scientists believe that the universe began to expand?
- A** During the singular event which occurred about 6 000 years ago
 - B** After the singular event which occurred 10–20 thousand million years ago
 - C** At the time the Hubble Telescope was launched into space
 - D** After the singular event which occurred 10–20 million years ago
- 8** In 1927, Georges Lemaître developed a hypothesis about the origin of the universe suggesting a changing radius to the size of the universe. His hypothesis was thorough and valid because:
- A** Lemaître worked with astronomers and made his theory in accord with observations, and had consequences which could be tested.
 - B** Lemaître did not use mathematical solutions but only used observations through a telescope.
 - C** Lemaître was the first to devise the theory.
 - D** Lemaître developed his hypothesis of the expanding atom from the writings of earlier philosophers.
- 9** Which statement about waves is true?
- A** Long waves have high frequency.
 - B** Short waves have high frequency.
 - C** Low frequency waves are the most dangerous.
 - D** Low frequency waves penetrate lead.

- 10** Astronomers today use every part of the electromagnetic spectrum as shown in the diagram below when they are making their observations.



Which part of the EMS did astronomers use until the twentieth century?

- A** Short wave radio
- B** Microwaves
- C** Visible light
- D** Gamma rays

Questions 11–13 refer to the following information. Originally, it was difficult to understand the cosmic background even though it was predicted before it was discovered. The astronomers Arno Penzias and Robert Wilson thought at first they had pigeon debris on their 6-metre horn antenna, but soon realised they were receiving the waves they were looking for in their investigation.

- 11** What piece of equipment were they using?
- A** A particularly sensitive radio telescope
 - B** An amplifier tuned to cosmic waves
 - C** A 10-metre refracting telescope
 - D** A cosmic background receiver

12 What does this radiation tell us about space?

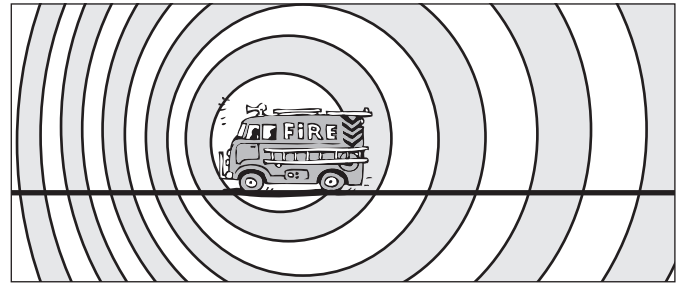
- ☐ A There was a 'big bang' that left radiation remnants.
- ☐ B All stars give off radio waves.
- ☐ C Galaxies are expanding.
- ☐ D Space is black because of the cosmic background.

13 This observation 50 years ago helped astronomers to understand more about the origin of the universe, as it has:

- ☐ A enabled astronomers to look at the universe with telescopes.
- ☐ B provided evidence that the universe began with a 'big bang'.
- ☐ C provided evidence that the universe began as a galaxy.
- ☐ D provided evidence that the universe began as a solar system.

14 Absolute zero on the Kelvin scale is equal to -273.15°C . During the Big Bang, a temperature of 3000 kelvin was likely. Now, after approximately 13 billion years, there is a thermal radiation temperature of 3 kelvin. What has probably caused this change?

- ☐ A The apparent difference in our measurement methods
- ☐ B Heat has been used up in the formation of stars.
- ☐ C The radiation has travelled freely through space.
- ☐ D Heat has been lost to other regions.



15 The Doppler effect as illustrated above shows waves bunching up and becoming higher in pitch as the fire engine approaches, and spreading apart and becoming lower in pitch as it moves away. In the electromagnetic spectrum, we see this as a:

- ☐ A blue shift as the noise from stars comes closer.
- ☐ B red shift as the noise from stars comes closer.
- ☐ C blue shift as light moves away.
- ☐ D red shift as light moves away.

16 Which explanation leads to the deduction that the Universe is expanding?

- ☐ A Blue light from the stars has a much shorter wavelength than red light.
- ☐ B Yellow light from the Sun is moving quickly.
- ☐ C Light from distant stars shows a red shift in relation to the Earth.
- ☐ D Green light has a short wavelength.

17 The PULSE@Parkes project allows students to control the famous Parkes radio telescope. With information about pulsars, you could help astronomers:

- ☐ A find the distances to these pulsars.
- ☐ B take optical photographs of pulsars.
- ☐ C calculate the age of the Milky Way.
- ☐ D look for extra-terrestrial life.

- 1|8** Deep space pictures from the Hubble satellite show objects further away than anything seen before. To achieve this, astronomers pointed Hubble cameras into an area of the universe with few visible objects, yet their pictures show many galaxies. How is this possible?
- ☐ A Ultraviolet light from distant objects took a long time to reach the Hubble satellite.
 - ☐ B Stray objects moved by at random times when the shutter was open.
 - ☐ C The orbits of the galaxies only brought them into view occasionally.
 - ☐ D Infra-red and visible light was collected by the cameras over many weeks.
- 1|9** In 1912, Henrietta Leavitt, an American astronomer, was studying Cepheid variables. These are stars that oscillate in brightness in regular cycles. They are easy to recognise in distant galaxies and are valuable to astronomers. Which term is commonly used for these stars?
- ☐ A Cyclic indicators
 - ☐ B Time shifters
 - ☐ C Yard sticks
 - ☐ D Standard candles
- 2|0** In 1994, Wendy Freedman and her colleagues in California used the Hubble telescope to measure the age of the universe. The information they found gave an age of 8–12 billion years, which does not fit with the age of some stars. Astronomers now have to:
- ☐ A develop models of the cosmos that account for all new information.
 - ☐ B choose not to work with new data that does not fit current theories.
 - ☐ C change the big bang theory slightly.
 - ☐ D revert to using the Steady State theory.
- 2|1** The Shelton-Duhalde-Jones supernova of 1987 was recorded on a photographic plate in Chile and as a sighting by an amateur astronomer in New Zealand. Which part of the spectrum was used to gain this information about the exploding star?
- ☐ A Visible light
 - ☐ B Infra-red light
 - ☐ C Radio waves
 - ☐ D Gamma rays
- 2|2** Astronomers are able to find new information with better telescopes like NASA's heat-sensing Spitzer Space Telescope. Spitzer can detect red stars in faraway galaxies because it is sensitive to infra-red light. Scientists may find information that does not fit with accepted theories. If new information is discovered, what should scientists do?
- ☐ A Stay with the present theory.
 - ☐ B Completely change everything they write.
 - ☐ C Use more rigorous testing to find the answer.
 - ☐ D Leave out the data that does not agree with the accepted theory.
- 2|3** Rotating neutron stars that are radio sources and blink on and off at a constant frequency are called:
- ☐ A radio stars.
 - ☐ B pulsars.
 - ☐ C neutrinos.
 - ☐ D black holes.

1 Edwin Powell Hubble made some major discoveries in the 1920s which led to the development of:

- A** the largest refracting telescope in the world.
- B** theories about the size of the universe and that the universe is expanding.
- C** the understanding of the various particles in space.
- D** the explanation of the formation of black holes and their place in the universe.

Questions 2 and 3 use the following information.

In Sesame Street, the characters explain the concept of near and far to young viewers by showing that a fly on a child's nose is close and looks big, while the same fly 5 metres away looks small. Astronomers use the same concept to tell distances in space, using Cepheid variable stars as their 'flies'.

2 Which characteristic of Cepheids allows them to be used for distance measurements?

- A** Many Cepheid stars have identical brightness cycles.
- B** All Cepheid stars are the same size and are at the same distance.
- C** All Cepheid stars are in the same area of the universe.
- D** Small Cepheid stars are much further away.

3 Cepheid stars pulsate because:

- A** they are so far away.
- B** there are two stars orbiting each other and one blocks the view of the other.
- C** they have used up their main supply of hydrogen fuel.
- D** all stars pulsate.

4 In the 1930s, nebulae were thought of as mysterious objects in the furthest reaches of the sky. What are they now known to be?

- A** Clusters of galaxies beyond our galaxy
- B** Clusters of stars in our galaxy
- C** Collections of dust and gas in our solar system
- D** Collections of dust and gas in our galaxy

5 Using present calculations, what is the most likely distance to the furthest clusters of galaxies in our universe?

- A** Hundreds of light years
- B** A few thousand light years
- C** Millions of light years
- D** Billions of light years

6 Galaxies proved to be so numerous that they needed to be classified. The common system in use is the:

- A** Linnean system, using 10 categories like L21, the Southern Cross galaxy.
- B** Hubble system, using size only.
- C** Hale-Bop system, using proximity to Earth.
- D** Messier system of basic galaxy shapes, like M31, the Andromeda galaxy.

7 For a measure of distance to faraway galaxies, scientists commonly use the red shift of an object instead of calculating its distance in light years because:

- A** the huge amounts of time involved means that the light has diminished.
- B** it is an easier figure to work with.
- C** the distances in light years cannot be measured.
- D** light varies in its speed, depending on the medium that it travels through.

8 Why aren't Astronomical Units used to measure distances to far galaxies?

- ☐ A One unit is only the distance from the Earth to the Sun.
- ☐ B The unit is too large for the distances involved.
- ☐ C More precision can be obtained by using kilometres.
- ☐ D The distance measurement also needs to allow for time measurements.

9 Which of the following are used in remote sensing?

- ☐ A Ultraviolet, X-rays, electromagnetic waves
- ☐ B FM waves, X-rays, pulsars
- ☐ C Ultraviolet, quasars, FM waves
- ☐ D Pulsars, X-rays, electromagnetic waves

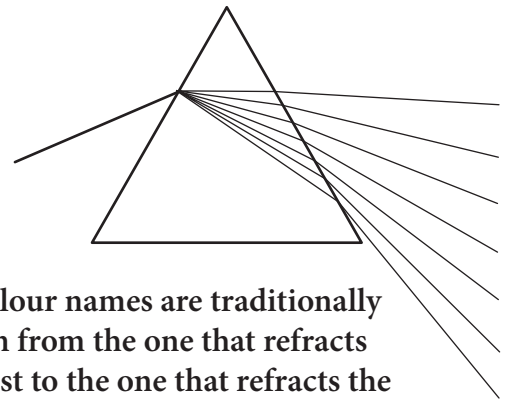
10 When people look through very good telescopes, which choice best describes the range of objects that can be seen?

- ☐ A Only objects in the solar system
- ☐ B Objects in the solar system and the nearest stars
- ☐ C Objects in the solar system and stars in our galaxy
- ☐ D Objects in the solar system, stars in our galaxy and in other galaxies

11 Which devices give astronomers better precision regarding celestial objects and the universe?

- ☐ A Satellites and light telescopes
- ☐ B Light telescopes and microscopes
- ☐ C Binoculars and space stations
- ☐ D Infra-red telescopes and radio telescopes

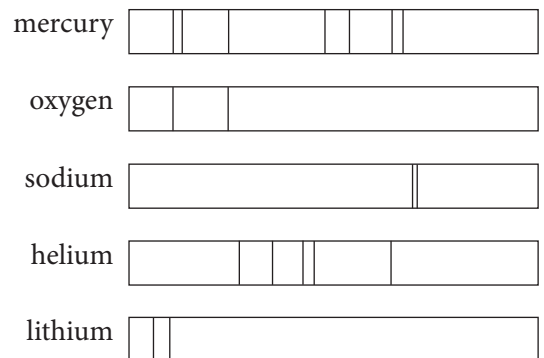
12 The spectroscope breaks white light into the colours of the rainbow.



The colour names are traditionally written from the one that refracts the least to the one that refracts the most, and are:

- ☐ A violet, blue, cyan, green, yellow, orange, red.
- ☐ B red, yellow, pink, green, orange, purple, blue.
- ☐ C red, orange, yellow, green, blue, indigo, violet.
- ☐ D red, yellow, blue.

13 Spectroscopy is a method for determining which elements are present in stars. Each element produces its own pattern of emitted light. This pattern is called a spectrum. The spectra of some elements are shown below.



A star has the following spectrum:



Which elements are present in the star, as shown by the spectrum?

- ☐ A Helium and mercury
- ☐ B Sodium and mercury
- ☐ C Helium and sodium
- ☐ D Oxygen and helium

14 Emission spectra provide information about the:

- ☐ A chemical composition of stars and potential harmful emissions moving away from the star.
- ☐ B physical structure of the star.
- ☐ C chemical composition of the star and whether it is moving away from the observer.
- ☐ D physical and chemical composition of the star.

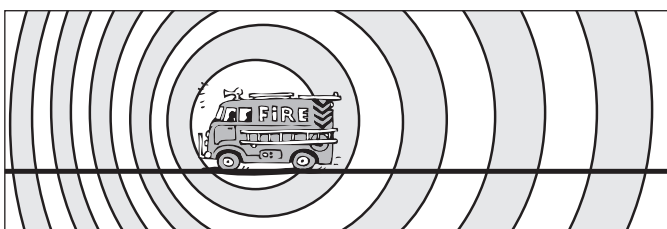
15 The Earth's atmosphere impedes most radiation coming from space and makes observations difficult. Which radiation types penetrate the atmosphere *most* easily?

- ☐ A Ultraviolet and X-rays
- ☐ B The visible and ultraviolet parts of the spectrum
- ☐ C Gamma rays and X-rays
- ☐ D The infra-red and visible parts of the spectrum

16 To observe other parts of the spectrum, astronomers must:

- ☐ A use satellite observatories.
- ☐ B build larger telescopes.
- ☐ C build observatories in cities with better power supplies.
- ☐ D use more computers attached to the telescopes.

Questions 17 and 18 use the following information.
The Doppler effect is used extensively in astronomy because it allows us to 'see' movement when objects are too far away to *really* see them. When Hubble saw the light from stars in distant galaxies, he saw that the spectral lines matched those of many familiar elements, but also showed a consistent red shift.



17 We experience the Doppler effect when a fire engine siren approaches us and then moves away from us. As the siren approaches, the waves bunch up and we hear a:

- ☐ A higher pitch.
- ☐ B lower pitch.
- ☐ C louder noise.
- ☐ D softer noise.

18 In the visible colour spectrum, the waves with the longest wavelength are:

- ☐ A blue.
- ☐ B green.
- ☐ C pink.
- ☐ D red.

19 What do astronomers think is the cause of the lines in the spectra of elements having a red shift?

- ☐ A Interference from the Earth's atmosphere causes the red glow.
- ☐ B As stars are moving away from Earth, the wavelength changes.
- ☐ C The heat from stars causes the red lines.
- ☐ D Galaxies are collapsing, causing the red shift in the wavelengths.

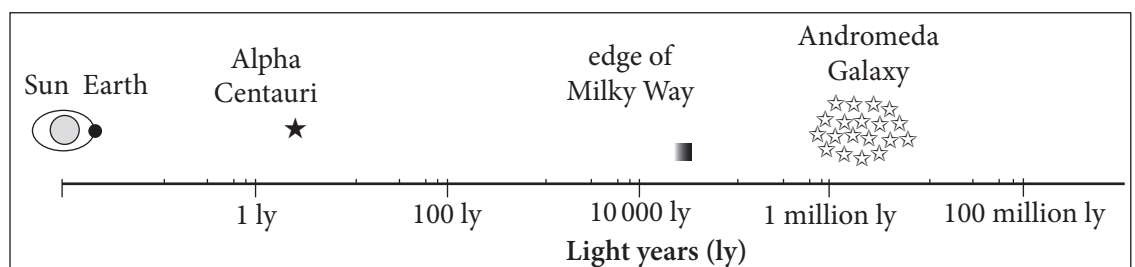
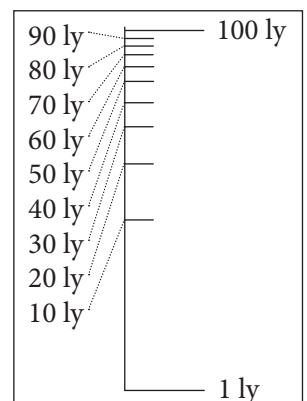
20 The European Council's Nuclear Research CERN facility in Geneva includes the Large Hadron Collider (LHC). Physicists and astronomers are hoping that it can be used to find out:

- ☐ A how far away the most distant planets really are.
- ☐ B new and profound insights into the workings of our universe.
- ☐ C how black holes have shaped our universe.
- ☐ D if there are extra-terrestrial beings in the universe.

- 1** Quasars (quasi-stellar radio sources) are some of the most distant objects that astronomers observe. They are incredibly far away, emitting large amounts of light in the centres of young galaxies. We are seeing them as they were billions of years ago. What can we infer from this observation?
- A** The galaxies themselves are evolving.
 - B** Distant galaxies are small.
 - C** The galaxies are running out of energy.
 - D** New galaxies are forming today in the outer reaches of space.
- 2** Black holes are likely to be in the centre of most galaxies and are energetically taking in gas. They have enormous gravitational pull and can:
- A** create new galaxies.
 - B** bend the path of light.
 - C** draw in remote galaxies.
 - D** lead to time warps.
- 3** The Milky Way is apparently moving forward in a huge area of space, pushing cosmic microwaves ahead of it like a ship pushing into waves in the ocean. This creates a slightly hotter area in front of our galaxy and a cooler area behind the galaxy, called a dipole anisotropy. This is another example of the:
- A** Doppler effect.
 - B** Scale diagram of space.
 - C** Hertzsprung-Russell star chart.
 - D** temperature probes on satellites.
- 4** The Cosmic Background Explorer COBE satellite gathered information for two years. This information was analysed by George Smoot and John Mather, who received the Nobel Prize in Physics in 2006 for their work. What was special about their project?
- A** This work backed up the big bang theory of the universe.
 - B** Their research showed life could be possible on planets in other galaxies.
 - C** The Pancake theory of galaxy clusters was developed from their work.
 - D** Smoot and Mather showed that space probes could go into the cosmic background.
- 5** The Great Attractor is the largest structure known in the universe and it correlates with the direction in which the Milky Way is moving. It is most likely to be a:
- A** completely new universe.
 - B** large group of supernova.
 - C** large concentration of galaxy clusters.
 - D** group of black holes.

Questions 6, 7 and 8 use the following information.

The scale of space is very difficult to draw on a page. The two diagrams show one way of representing the distances involved.



6 Using the two scales, what is the approximate distance from the Sun to the edge of the Milky Way?

- A** 100 light years
- B** 2 000 light years
- C** 50 000 light years
- D** 1 000 000 light years

7 Using the two scales, what is the approximate distance from the Sun to Alpha Centauri?

- A** 4 light years **B** 10 light years
- C** 50 light years **D** 100 light years

8 When measurements become huge, the method of showing scale may need to change. Two types of scales are a linear scale in which the divisions are uniformly spaced and a non-linear scale where they are not evenly spaced. Which is the best way to describe the scale of space objects drawn on the previous page?

- A** A non-linear scale with each major division 100 times more compressed than the previous one
- B** A non-linear scale with each division the square of the previous one
- C** A linear scale with each major division equally spaced but compressed across the page
- D** A linear scale with each division showing a geometric progression

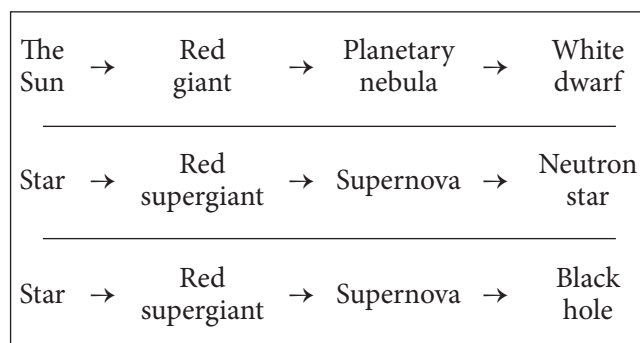
9 Stars have life cycles caused by gravitational forces and nuclear reactions. Which characteristics of a star change during its development?

- A** Only the size of the star
- B** The colour, surface temperature and diameter
- C** The colour and diameter only
- D** Surface temperature and gravitational pull

10 What is the main fuel for fusion reactions in stars?

- A** Carbon
- B** Helium
- C** Hydrogen
- D** Oxygen

11 The diagram shows three different stages of three types of stars.



Which of these statements is correct?

- A** Our Sun will eventually become either a black hole or a neutron star.
- B** All supergiants will become black holes.
- C** All stars will eventually explode and become supernova.
- D** A supergiant may become either a black hole or a neutron star.

12 Stars change as they grow older, but there are different pathways depending on their initial size. Which statement best describes the possible future of some stars?

- A** A dying star may go 'supernova'.
- B** As stars get older, they slowly and steadily become cooler.
- C** Stars become black holes in their final stage.
- D** Red stars show that the surface temperature is becoming hotter as they age.

13 Over time, astronomers have developed various theories and models about the stars and planets. However, they have recently formulated models that describe a star's life cycle. Why has it taken so long to develop star cycle models?

- ☐ A Astronomers were too busy developing the theory of evolution.
- ☐ B Astronomers were not able to collect relevant data about stars.
- ☐ C The light from stars takes a very long time to reach Earth.
- ☐ D The universe began billions of years ago and so stars take a long time to evolve.

14 Star formation occurs because of:

- ☐ A an attractive gravitational force on regions of gas.
- ☐ B gravitational pull from other stars.
- ☐ C the protons in the gas igniting spontaneously.
- ☐ D the gravity of the protons inducing electron gravitational pull.

15 Energy is produced in the Sun as a result of:

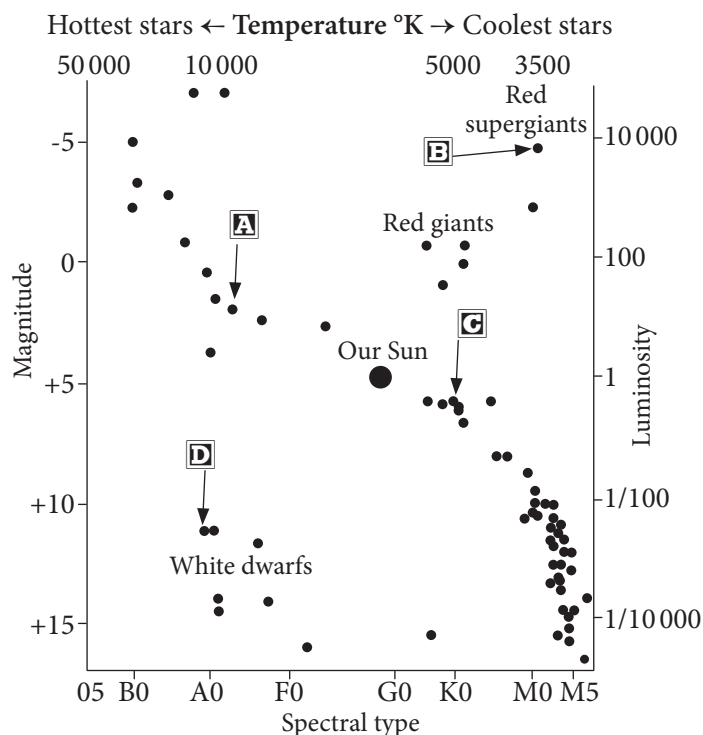
- ☐ A nuclear fusion.
- ☐ B nuclear fission.
- ☐ C neutron decay.
- ☐ D proton decay.

16 During the life cycle of a star, the hydrogen becomes exhausted and the star begins to fuse helium into carbon and puffs up its atmosphere. This will cause it to become a:

- ☐ A giant planet.
- ☐ B black carbon hole.
- ☐ C supernova.
- ☐ D red giant.

Questions 17 and 18 use the following information.

The Hertzsprung-Russell diagram of the stars is shown in the next column. The horizontal axis shows spectral type and temperature, from the hottest stars on the left to the coolest on the right. The vertical axis shows the luminosity of the stars, with those 10 000 times brighter than the Sun at the top, and those only 1/10 000th of its brightness at the bottom.



17 Which letter on the diagram represents Sirius, an extremely hot star but much less luminous than the Sun?

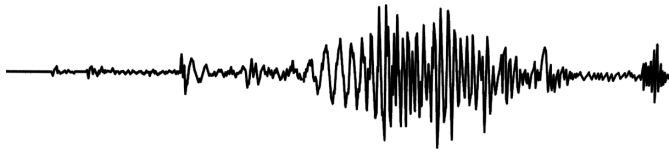
- ☐ A
- ☐ B
- ☐ C
- ☐ D

18 Using the information in the diagram, which feature is often common to the most luminous stars?

- ☐ A They are giant stars.
- ☐ B A star's age
- ☐ C Temperature in the star
- ☐ D Spectral type

19 After the star has lost its outer material, it will develop an expanding nebulous area and then it may simply become a white dwarf. The end fate of a star depends on:

- ☐ A the initial size of the star.
- ☐ B the position of the star in the universe.
- ☐ C the length of time since its origin.
- ☐ D the distance from it to the centre of the galaxy.



1 The image above, which shows the occurrence of an earthquake, is called:

- ☐ A a seismogram.
- ☐ B a seismic wave.
- ☐ C a seismometer.
- ☐ D an epicentre.

2 The term Pangaea describes:

- ☐ A the Aboriginal name for Australia.
- ☐ B the name of a supercontinent that included only Australia, Antarctica and South America.
- ☐ C a continent formed by continental drift.
- ☐ D all the continents joined together in one large land mass.

3 The name Rodinia belongs to:

- ☐ A a subcontinent of Antarctica.
- ☐ B a super continent that formed before Pangaea.
- ☐ C an island arc in the Pacific Ocean.
- ☐ D a volcanic crater near Mt Kilimanjaro.

4 Continental drift:

- ☐ A is a theory based on the idea that tectonic plates actually float on a sea of semi-fluid rock.
- ☐ B means continents are literally floating in the oceans.
- ☐ C is responsible for naturally occurring phenomena such as volcanic activity.
- ☐ D is evidence that the Mid Atlantic Ridge exists.

5 Although nobody has seen the inside of the Earth, scientists believe they have a good idea of what it looks like. This is because:

- ☐ A they have been able to use powerful drills to reach the centre of the Earth.
- ☐ B there are 'windows' into the Earth such as earthquakes and volcanic activity.
- ☐ C they have been able to travel into the Earth via natural tunnels.
- ☐ D remote sensing from space has been used to see into the Earth.

6 A tsunami is caused by the effect of:

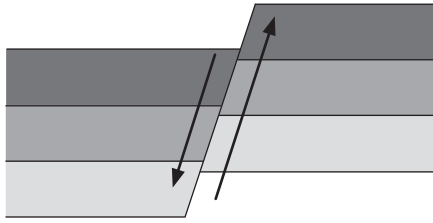
- ☐ A a large wave at high tide.
- ☐ B a series of waves caused by underwater earthquakes or volcanic eruptions.
- ☐ C the aftershock that follows an earthquake.
- ☐ D the epicentre of an earthquake.

7 The Himalayan Mountains are the result of:

- ☐ A sedimentary rock buckling due to volcanic activity.
- ☐ B volcanic eruptions in India 350 million years ago.
- ☐ C eroded rock material.
- ☐ D tectonic plate collision and uplift.

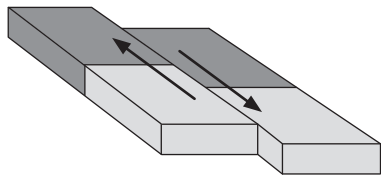
8 Three variations in the properties of the Earth's crust that geophysicists can measure when searching for ore bodies are:

- ☐ A gravity, magnetism and radioactivity.
- ☐ B pressure, temperature and mineral abundance.
- ☐ C electrical conductivity, altitude and depth.
- ☐ D igneous activity, seismic activity and groundwater.



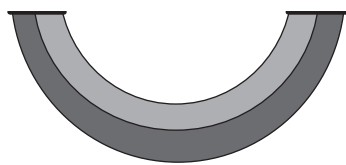
9 The diagram above shows:

- ☐ A a normal fault.
- ☐ B a reverse fault.
- ☐ C an anticline.
- ☐ D a syncline.



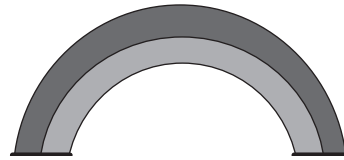
10 The diagram above shows:

- ☐ A a normal fault.
- ☐ B a reverse fault.
- ☐ C a strike slip fault.
- ☐ D a convergent boundary.



11 The diagram above shows:

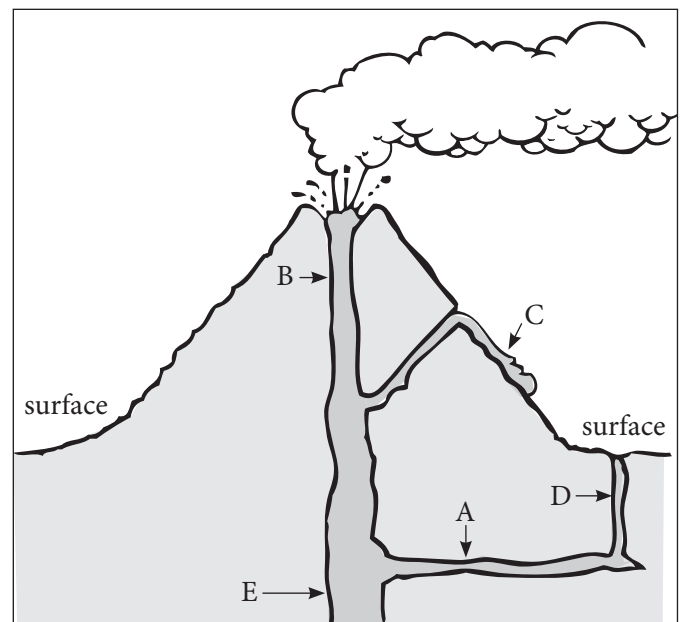
- ☐ A an anticline.
- ☐ B a reverse fault.
- ☐ C a syncline.
- ☐ D the result of weathering.



12 The diagram above shows:

- ☐ A the result of erosion deposition.
- ☐ B an anticline.
- ☐ C a syncline.
- ☐ D a fault line.

Questions 13–16 refer to the following diagram.



13 Which letter represents the volcano's vent?

- ☐ A A
- ☐ B B
- ☐ C C
- ☐ D D

14 Which letter represents a dyke?

- ☐ A A
- ☐ B B
- ☐ C C
- ☐ D D

15 Which letter represents a sill?

- ☐ A
- ☐ B
- ☐ C
- ☐ D

16 Which letter represents the lava flow?

- ☐ A
- ☐ B
- ☐ C
- ☐ D

17 Areas subject to earthquakes are also prone to volcanic activity because:

- ☐ the reason is still not known.
- ☐ volcanoes are formed following large earthquakes.
- ☐ these areas correspond to weaknesses in the Earth's crust.
- ☐ volcanic eruptions are the only cause of earthquakes.



18 The diagram above is a representation of which super continent?

- ☐ Laurasia
- ☐ Pangaea
- ☐ Gondwana
- ☐ Columbia

19 Which of the following is evidence that indicates that super continents broke into smaller continents?

- ☐ The dinosaurs vanished.
- ☐ The sea floor is spreading.
- ☐ The same rock types are on adjoining continental margins.
- ☐ Eucalyptus leaves forming coal seams are present in every continent.

20 The process of subduction relates to which three factors?

- ☐ Continental collision, vulcanism and gravity
- ☐ Continental collision, earthquakes and rock density
- ☐ Dip and strike of rocks, gravity and rock density
- ☐ Convection currents, gravity and rock density

21 The Australian tectonic plate is moving north-north-east by approximately 2 centimetres each year. The effects of this movement cause:

- ☐ flooding in the Philippines.
- ☐ synclines to form off the coast of China.
- ☐ subduction of the Philippine plate.
- ☐ earthquakes in the oceans near countries north of Australia.

22 The epicentre of the Newcastle earthquake in 1989 was situated 12 kilometres underground. The earthquake had a magnitude of 5.6 and produced shear waves. This earthquake could be classified as being:

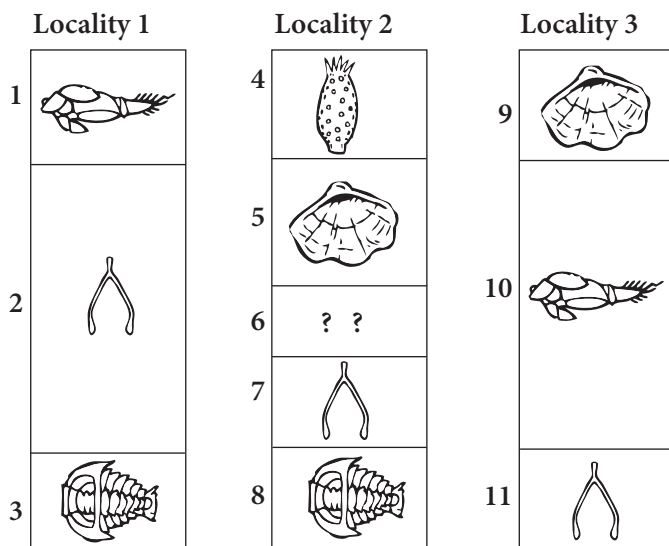
- ☐ very deep.
- ☐ shallow.
- ☐ deep.
- ☐ extremely deep.

- 1** Most fossils are found in:
- A** rock layers that were formed originally under water.
 - B** rocks formed by cooling lava.
 - C** soils.
 - D** waters of seas and lakes.
- 2** Which of the following shows more evidence of fossilisation than the other choices?
- A** The re-mineralised shell of an animal that lived in water
 - B** Any bone, tooth or shell used by early man
 - C** Any animal or plant that lived in ancient times, but no longer exists
 - D** The archaeological remains of campfires
- 3** The Flintstone family have a pet dinosaur. This is impossible because:
- A** dinosaurs don't make good pets.
 - B** dinosaurs evolved a long time before early man.
 - C** dinosaurs became extinct just as humankind was evolving.
 - D** dinosaurs became extinct at least 65 million years before humankind evolved.
- 4** The dinosaurs most likely became extinct because:
- A** they had just finished evolving.
 - B** vulcanism caused a dramatic climate change.
 - C** mammals began to replace the dinosaurs.
 - D** sea levels rose.
- 5** A rock layer contains fossils of clam and oyster shells. The kind of environment that existed at the time these fossils were formed was a:
- A** forest.
 - B** swamp.
 - C** shoreline.
 - D** river waterfall.
- 6** Soft-bodied animals such as jellyfish are not generally found as fossils because:
- A** the rocks squash them.
 - B** environmental conditions when they die harden their bodies.
 - C** environmental conditions when they die decompose their bodies.
 - D** mud silts are poor preserving agents.
- 7** Trilobites were common and abundant from the Cambrian to the Permian periods. Trilobites are good indicator fossils because:
- A** they are only found in one type of rock.
 - B** specific types lived in each period in a small locality.
 - C** specific types lived in each period over a wide area.
 - D** each type of trilobite has the same features.
- 8** Relative dating of fossils can be done by using:
- A** knowledge of rock strata and its geology.
 - B** radiometric dating.
 - C** bioluminescent dating.
 - D** the placement of igneous intrusions in the strata.

9 The main group of back-boned animals that lived in the waters of the Devonian period were:

- ☐ A pterosaurs.
- ☐ B fish.
- ☐ C mammals.
- ☐ D amphibians.

Questions 10–13 refer to the following diagram.



10 Fossils have been identified in each rock layer and locality. Some fossils were common in two or all three locations. How many different types of fossils were identified?

- ☐ A 4
- ☐ B 5
- ☐ C 7
- ☐ D 11

11 Which fossils are common to all localities?

- ☐ A 1, 6, 10
- ☐ B ?, 5, 9
- ☐ C 2, 7, 11
- ☐ D 1, 10, ?

12 Which number indicates the youngest fossil?

- ☐ A 5
- ☐ B 4
- ☐ C 1
- ☐ D 3

13 Which rock layers in the three different localities are the same age?

- ☐ A Rock layers 2, 5 and 9
- ☐ B Rock layers 2, 7 and 11
- ☐ C Rock layers 3, 8 and 11
- ☐ D Rock layers 1, 7 and 9

14 A mammal fossil is supposedly found in a coal seam. This is unlikely because:

- ☐ A most coal seams were formed approximately 250 million years ago or more.
- ☐ B mammal fossils are found only in shale type rocks.
- ☐ C mammals only started to evolve approximately 17 million years ago.
- ☐ D fossils do not preserve well in coal seams.

15 The Ginkgo tree evolved in the Permian period and still exists today. This is because:

- ☐ A the Ginkgo has not been able to adapt to the climate as it changed.
- ☐ B conditions in some places today are similar to those in the Permian period.
- ☐ C it is not a living fossil.
- ☐ D its seeds take a long time to germinate.

- 16** Refer to the following table to answer this question.

Million of years ago	Climate
2–62	Temperate to dry
160–275	Tropical/ice ages
300–400	Warm to tropical
450–600	Wet and cloudy

Dinosaurs would have used the Sun's heat to maintain their body temperature, and living reptiles still use it today. Dinosaurs died out about 160 million years ago. They probably did not evolve again because:

- ☐ A no viable eggs were left.
- ☐ B plants did not evolve again.
- ☐ C climatic changes were unsuitable for dinosaurs.
- ☐ D there was not enough water on the Earth.

- 17** Over many years, the original bone or shell is sometimes replaced by minerals from surrounding rock. This process is called:

- ☐ A sedimentation.
- ☐ B fossilisation.
- ☐ C crystal formation.
- ☐ D metamorphism.

- 18** Fossil formation requires several stages. The correct fossilisation sequence is:

- ☐ A mineralisation, rapid burial, replacement.
- ☐ B rapid burial, replacement, mineralisation.
- ☐ C replacement, mineralisation, rapid burial.
- ☐ D slow burial, replacement, crystallisation.

- 19** Ice ages, vulcanism and extra-terrestrial impacts have all been major events contributing to the extinction of organisms. Which geological period or boundary between periods saw the extinction of the dinosaurs?

- ☐ A Permian–Devonian
- ☐ B Devonian
- ☐ C Cretaceous–Tertiary
- ☐ D Ordovician–Silurian

- 20** Not all extinctions are due to catastrophic events like vulcanism. The extinction of the Tasmanian Tiger was probably due to:

- ☐ A a bacterial disease.
- ☐ B an unknown virus.
- ☐ C it being shot by farmers.
- ☐ D it being eaten by Tasmanian Devils.

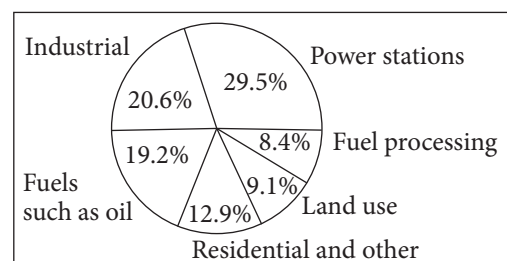
- 21** Australian megafauna became extinct because of:

- ☐ A disease.
- ☐ B fire.
- ☐ C the introduction of foxes.
- ☐ D ecosystem alteration.

- 22** Many Australian animal and plant species are in danger of becoming extinct. The main reason for this is because:

- ☐ A there are too many people in Australia using up too many resources.
- ☐ B genetically modified plants are causing the extinctions.
- ☐ C feral animals such as dogs, cats and cloven-hoofed animals prey on them for food.
- ☐ D there is too much land clearing near the coast.

- 1** Fossil fuels are made from three chemical elements. Two of the elements in these fuels are:
- A** sulphur and nitrogen
 - B** carbon and hydrogen
 - C** carbon and nitrogen
 - D** nitrogen and hydrogen
- 2** Which organisms make oil and gas?
- A** Plankton
 - B** Large plants
 - C** Large, shelled animals
 - D** Flotsam
- 3** Which organisms make coal?
- A** Very small marine plants and animals
 - B** Terrestrial animals
 - C** Land plants
 - D** Cephalopods and ammonites
- 4** Which gases are produced when fossil fuels are combusted?
- A** Carbon dioxide, hydrogen, methane, hydrogen sulphide
 - B** Carbon dioxide, hydrogen sulphide, nitrogen oxides, methane
 - C** Carbon dioxide, methane pentahydride, hydrogen sulphide, hydrogen
 - D** Carbon dioxide, propane, methane pentahydride, ethane
- 5** The gas used to combust fossil fuels is:
- A** oxygen dichloride.
 - B** oxygen.
 - C** ozone.
 - D** oxygen trifluoride.
- 6** A greenhouse gas is best defined as any gas:
- A** that absorbs ultraviolet radiation.
 - B** that absorbs cosmic ray radiation.
 - C** that is found only in greenhouses.
 - D** that absorbs infra-red radiation.
- 7** Greenhouse gases heat the atmosphere and Earth's surface by which process?
- A** Absorption of all radiated heat
 - B** Absorption and re-emission of some infra-red radiation
 - C** Re-emission of some ultraviolet radiation from the ozone layer
 - D** Re-emission of all gases in the atmosphere
- 8** Which is *not* a greenhouse gas?
- A** Carbon dioxide gas
 - B** Nitrogen gas
 - C** Carbon monoxide gas
 - D** Sulphur hexafluoride gas
- 9** Australia has 500 years' worth of coal resources that can be used domestically or exported. The pie chart represents emissions of carbon dioxide gas.



Source: www.cutco2.org/facts.php

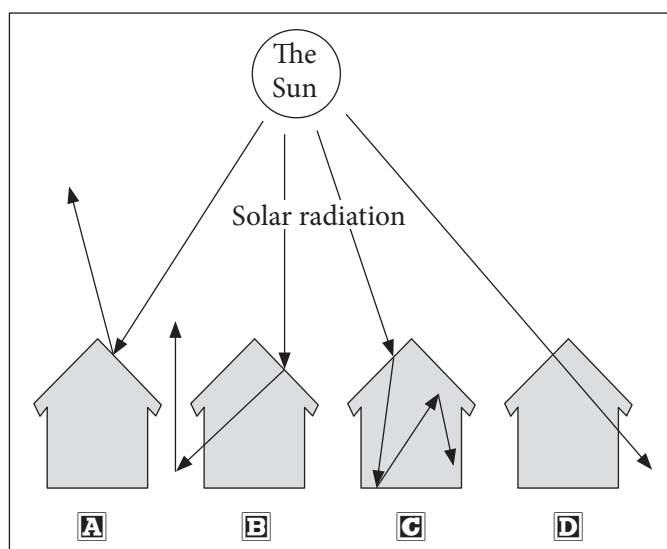
With renewable energy sources producing more energy, which of the following sources of emissions from the pie chart will most likely remain unchanged?

- A** Coal-fired power stations
- B** Fuels such as oil
- C** Land use
- D** Industrial use

10 When combusted, diesel fuel like petrol produces a variety of gases and very small particulate matter. Two of these solid substances are:

- A** carbon and sulphur dioxide.
- B** carbon dioxide and sulphur.
- C** nitrogen oxide and sulfur trioxide.
- D** carbon and sulphur.

11 The diagram below shows four models of the Greenhouse Effect. Which is the correct model for showing the infra-red effect?



12 The enhanced Greenhouse Effect occurs in which level of the atmosphere?

- A** The highest level of the atmosphere
- B** The middle level of the atmosphere
- C** The lowest level of the atmosphere
- D** The whole atmosphere

13 Over the past 250 years, carbon dioxide levels in the atmosphere have increased by 25%. The main cause of this increase is:

- A** agricultural production.
- B** horticultural production.
- C** industrialisation and energy production.
- D** land clearing for cities and agriculture.

14 The main greenhouse gas, carbon dioxide, increased to 379 ppm between 1995 and 2005. It is anticipated that fossil fuel emissions will increase by 45% between 2005 and 2015, bringing the level of carbon dioxide to a higher level. The expected level is most likely to be:

- A** 390 ppm.
- B** 420 ppm.
- C** 550 ppm.
- D** 470 ppm.

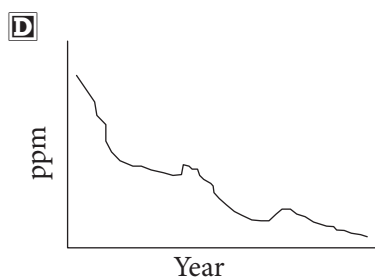
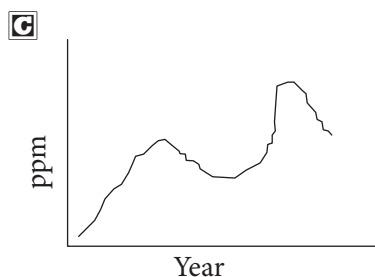
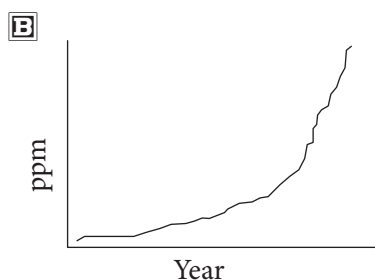
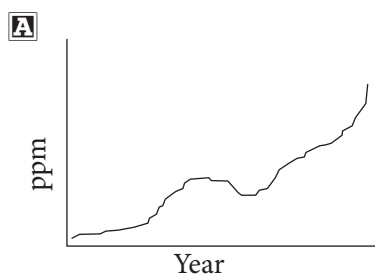
15 Below are four simplified chemical reactions that relate to the burning of fossil fuels. Which one shows the correct complete combustion reaction?

- A** petrol + oxygen → carbon dioxide + sulphur trioxide
- B** petrol + oxygen → nitrogen gas + carbon dioxide
- C** petrol + oxygen → carbon monoxide + water
- D** petrol + oxygen → carbon dioxide + water

16 Nitrous oxide in the atmosphere is about 15% higher now than it was 200 years ago. This increase is due to:

- A** industrial emissions, nitrogen fertilisers and bushfires.
- B** bushfires, landfill recycling and motor vehicle exhausts.
- C** motor vehicle exhausts, industrial emissions and nitrogen fertilisers.
- D** nitrogen fertilisers, bushfires and landfill recycling.

17 Which graph best indicates the increase of greenhouse gases since the Industrial Revolution?



18 Nitrous oxide is emitted from fossil fuels and reacts with other gases. One gas in the atmosphere prevents UV light penetration and the other is a greenhouse gas. The two gases are:

- A** hydrogen and ozone.
- B** water vapour and sulphur dioxide.
- C** ozone and water vapour.
- D** water vapour and carbon dioxide.

19 It has been demonstrated that ozone production is accelerating and causing particulates to remain in the air, contributing to heart disease. The fine soot is emitted from vehicles and machines. This soot comes from which fossil fuel?

- A** Natural gas combustion
- B** Methane combustion
- C** Diesel combustion
- D** Oil combustion

20 When natural gas (methane) is completely combusted, it produces two products. These products are:

- A** carbon and oxygen gas.
- B** carbon dioxide and water vapour.
- C** carbon and water vapour.
- D** carbon and hydrogen gas.

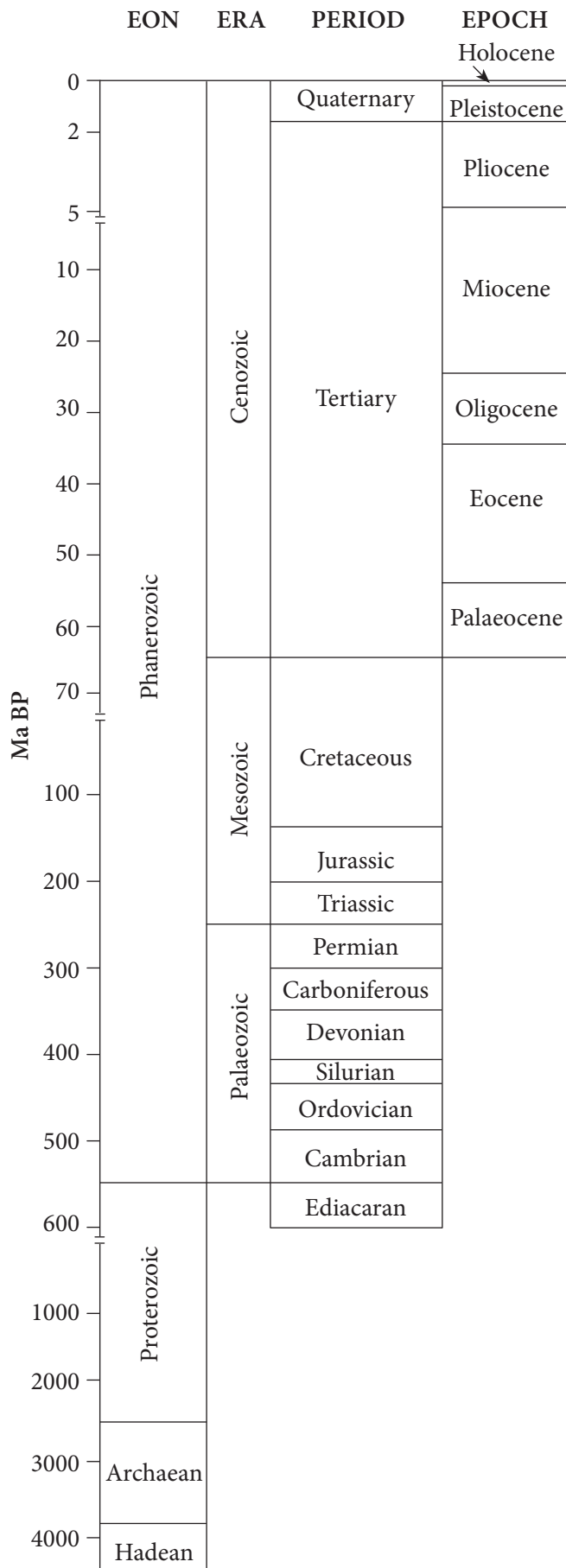
21 Which of the following is not a major greenhouse gas?

- A** carbon dioxide
- B** methane
- C** oxygen
- D** water vapour

22 When ChloroFluoroCarbons (CFCs) react with ozone, they destroy the ozone molecules and leave chlorine atoms as well as oxygen molecules in the stratosphere. The reaction is a multi-step process. The correct final reaction is:

- A** $\text{ClO} + \text{O}_3 \rightarrow \text{Cl} + 2\text{O}_2$
- B** $\text{ClO} + \text{O}_3 \rightarrow \text{Cl}_2 + 4\text{O}$
- C** $\text{ClO} + \text{O}_3 \rightarrow 2\text{Cl} + \text{O}_2$
- D** $\text{ClO} + \text{O}_3 \rightarrow \text{Cl} + 4\text{O}$

The geological time scale

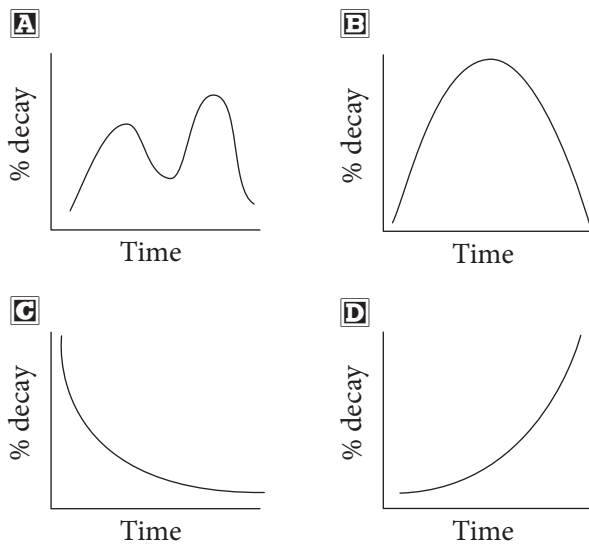


- The correct order of the geological time scale from the largest unit of time to the smallest unit of time is:
 - epoch, period, eon, era.
 - eon, era, period, epoch.
 - era, period, eon, epoch.
 - period, epoch, eon, era.
- Which of the following is in the correct order from oldest to youngest?
 - Cambrian, Ordovician, Silurian, Devonian
 - Cretaceous, Jurassic, Tertiary, Triassic
 - Devonian, Cretaceous, Carboniferous, Permian
 - Triassic, Palaeocene, Miocene, Oligocene
- The Principle of Superposition states that:
 - the sandiest layer is always on the bottom layer of rock.
 - the bottom layer is the youngest layer of rock.
 - the bottom layer is the oldest layer of rock.
 - the top layer is the oldest layer of rock.
- Fossils provide information and correlation about rock layers in chronological order on a global basis. Which fossil sequence is the correct one?
 - Velvet worms, dinosaurs, trilobites, brachiopods
 - Amphibians, fish, trilobites, brachiopods
 - Velvet worms, trilobites, fish, amphibians
 - Brachiopods, ammonites, dinosaurs, amphibians

- 5** There are several methods for determining the age of rocks. Which method is the most accurate for determining the age of very old rocks?

☐ A Bioluminescence
☐ B Relative dating
☐ C Index fossil markers
☐ D Radiometric dating

- 6** Dating by radiometric methods requires the use of isotope half-lives that decay over time into another element. Which graph is the correct one for radiometric dating?



- 7** You have been asked to date a rock that is thought to be 2 billion years old. Which isotope would you use to determine its age?

☐ A Nitrogen-16
☐ B Carbon-14
☐ C Beryllium-4
☐ D Potassium-40

- 8** Which isotope would you use to determine the age of the remains of a mammal that died 50 000 years ago?

☐ A Nitrogen-14
☐ B Carbon-14
☐ C Oxygen-16
☐ D Uranium-235

- 9** You have discovered four different rock layers. The bottom layer is shale, the next is conglomerate, the top layer is sandstone and the fourth rock type is basalt which cuts through the other three layers. Which rock type is the youngest?

☐ A Conglomerate
☐ B Basalt
☐ C Shale
☐ D Sandstone

- 10** Dinosaurs probably became extinct due to an asteroid impact and vulcanism. A layer of iridium determines the period in which they became extinct. This layer is found between which two periods?

☐ A Carboniferous–Permian
☐ B Miocene–Pliocene
☐ C Cretaceous–Tertiary
☐ D Triassic–Jurassic

- 11** The era in which upright apes and humans evolved is called the:

☐ A Mesozoic.
☐ B Archaeozoic.
☐ C Cenozoic.
☐ D Proterozoic.

- 12** Iron oxides (rust) did not exist several billion years ago, as limited amounts of oxygen were present in the atmosphere until the process of photosynthesis occurred. The eon in which limited oxygen was present is called:

☐ A the Hadean.
☐ B the Proterozoic.
☐ C the Ediacaran.
☐ D the Cambrian.

13 Western Australia has rocks that have been dated to 4.2 billion years old, while on the east coast of Australia some of the oldest rocks are 350–450 million years old. This indicates that Australia grew from west to east. Which of the following reflect the geological growth of Australia?

- ☐ A Silurian–Devonian
- ☐ B Ediacaran–Cambrian
- ☐ C Cambrian–Ordovician
- ☐ D Pre-Cambrian–Cretaceous

14 In the late 1500s it was thought that the world had been created 4004 years ago. The person who proclaimed that the world was created at that time was:

- ☐ A Bishop Ussher.
- ☐ B Albert Einstein.
- ☐ C Charles Lyell.
- ☐ D James Hutton.

15 Who was the founder of modern geology?

- ☐ A Hutton
- ☐ B Werner
- ☐ C Cuvier
- ☐ D Newton

16 Africa and South America were once joined together. The evidence for this comes from fossils dated at 250–270 million years ago. These fossils come from which period?

- ☐ A Pollens from the Quaternary
- ☐ B Reptile fossils from the Permian
- ☐ C Trilobite fossils from the Silurian
- ☐ D Fish fossils from the Devonian

17 The geological rock record indicates chemical, physical and biological changes over time. Biologically, the fossil record demonstrates that:

- ☐ A evolution is limited to only the Quaternary period.
- ☐ B evolution is an ongoing process.
- ☐ C evolution is limited to the Cambrian period only.
- ☐ D evolution doesn't occur.

18 Stromatolite fossils are cyano-bacterial mats that form in saline and still waters. Cyanobacteria were the first organisms to carry out photosynthesis. These organisms first produced oxygen:

- ☐ A 250 million years ago.
- ☐ B 25 million years ago.
- ☐ C 2.5 million years ago.
- ☐ D 2500 million years ago.

19 Stromatolite fossil mats in the Jenolan Caves were formed from water drips. This indicates that they formed from:

- ☐ A fresh water with dissolved salts and light.
- ☐ B turbulent estuarine water.
- ☐ C slow-moving salty water and no light.
- ☐ D turbulent salty water.

20 A fossil pelvis of an animal shows muscle attachment sites that indicate that it walked upright. The age of the rock is 2.5 million years. The pelvis is most likely to belong to which organism?

- ☐ A A chimpanzee
- ☐ B A Bonobo
- ☐ C An Australopithecine
- ☐ D A Homo sapiens

- 1** The removal of deep-rooted trees in certain farming areas causes the land to become infertile over time. This is because the removal of deep-rooted trees causes:
- ☐ A fertilisers to percolate into the aquifer.
 - ☐ B salt to rise to the surface.
 - ☐ C soil organisms to die.
 - ☐ D extensive water erosion.
- 2** The Great Barrier Reef is under threat from bleaching which is caused by:
- ☐ A oil pollution from passing ships.
 - ☐ B oxygen depletion due to climate change.
 - ☐ C insufficient nutrients after a cyclone.
 - ☐ D excess fertilisers flowing into the reef and increased water temperatures.
- 3** Excessive amounts of nutrients in Australian rivers and dams cause the depletion of oxygen in the warm water of summer. This is caused by:
- ☐ A insufficient algae in the water.
 - ☐ B algal blooms.
 - ☐ C a lack of biodiversity.
 - ☐ D insufficient organisms to consume the nutrients.
- 4** The cane toad, *Bufo marinus*, is an introduced animal that is devastating many native Australian animal populations. The reason for its introduction in Australia was to:
- ☐ A consume sugar-cane algae.
 - ☐ B remove bull ants from sugar farms.
 - ☐ C consume the cane beetle.
 - ☐ D stop white mice from eating young sugar-cane plants.
- 5** The fox is an introduced carnivore that consumes many native animals. The reason for its introduction to Australia in the 1800s was to:
- ☐ A remove the dingo from the bush.
 - ☐ B allow people to enjoy the sport of fox hunting.
 - ☐ C provide an animal that would eat rats.
 - ☐ D scavenge food scraps from urban areas.
- 6** Urban areas are subject to soil erosion due to the rapid run-off of water. The cause of this is:
- ☐ A large areas of flat, grassed playing fields.
 - ☐ B large areas of bitumen and concrete.
 - ☐ C insufficient water collection for reuse.
 - ☐ D poor maintenance of gardens and parks.
- 7** Which emissions and temperature cause the form of air pollution called brown haze?
- ☐ A Power station emissions from generating more electricity in the morning
 - ☐ B Household emissions and late night temperature drops
 - ☐ C Vehicle and factory emissions and low morning temperature
 - ☐ D Factory emissions and midday temperatures
- 8** Human use of chemicals in the food chain is resulting in the extinction of many plants and animals. This process is called:
- ☐ A bioremediation. ☐ B bioaccumulation.
 - ☐ C bio-reduction. ☐ D bio-oxidation.
- 9** Our atmosphere has been polluted by various manufactured gases. Which of the following gases is not a greenhouse gas?
- ☐ A Water vapour ☐ B Methane
 - ☐ C Nitrous oxide ☐ D Nitrogen

- 110** Chlorofluorocarbons were developed for use in refrigeration, air conditioners and aerosol propellants. These gases affect one layer in the atmosphere and allow harmful UV radiation to strike the Earth. Which gas layer do they affect?
- A** Methane **B** Argon
C Ozone **D** Hydrogen
- 111** Motor vehicles emit carbon dioxide, nitrogen oxides and other gases from the combustion of petrol and diesel. When these gases are combined with water vapour, they are most likely to form:
- A** solid carbon. **B** acid rain.
C alkali rain. **D** sulphurous acid.
- 112** Coal-fired power stations have hot water run-off that is detrimental to aquatic organisms because the hot water:
- A** increases aquatic bacteria.
B decreases dissolved oxygen.
C increases dissolved oxygen.
D decreases aquatic bacteria.
- 113** Many consumer garbage items that are used as landfill produce leached liquids. These liquids are often toxic and are disposed of by which method?
- A** Chemical degradation
B Physical separation
C Centrifugation
D Dumping in the environment
- 114** Deforestation and the removal of grasslands reduce the amount of oxygen available to all organisms. The process to make oxygen is known as:
- A** aerobic respiration.
B photosynthesis.
C transpiration.
D anaerobic respiration.
- 115** In aquatic systems such as oceans, many dolphins and turtles die because of human produced litter that has found its way out to sea. The highest death rate in dolphins and turtles is caused by:
- A** paper cups.
B wood shavings.
C plastic bags.
D cloth rags.
- 116** Since the Industrial Revolution, the world's oceans have become more acidic than in any previous period. The cause of acidification of the oceans is:
- A** nitrogen oxide emissions.
B oceanic waste disposal of detergents.
C effluent production.
D carbon dioxide emissions.
- 117** Many cities around the world have increased pollution levels and temperatures while rainfall has decreased. This is due to which main factor?
- A** Large-scale farming
B Thermal and particle pollution
C Increased mining activity
D Loss of vegetation
- 118** Many turtles along the Queensland coast lay their eggs inappropriately in front of brightly lit hotels. The reason for this is:
- A** insufficient sand on the beach.
B light pollution.
C nocturnal predators.
D not enough moonlight.

- 1** The naturally occurring substances from which many metals can be chemically extracted are called:
- ☐ A alloys.
 - ☐ B elements.
 - ☐ C ores.
 - ☐ D sedimentary rocks.
- 2** Gold is a very valuable metal that can be found in a free state. It is possible to find pure gold because:
- ☐ A gold is a very heavy metal.
 - ☐ B like most metals, it can be obtained free of impurities.
 - ☐ C gold is a very unreactive metal.
 - ☐ D gold is very chemically active.
- 3** Which energy source will most likely outlast other energy sources?
- ☐ A Solar heat
 - ☐ B Coal
 - ☐ C Petroleum
 - ☐ D Geothermal
- 4** There are three processes used in separating a metal from the rock in which it is found. These processes are listed below but not in the correct order.
- Refining: purifying the metal
 - Concentration: separation of the valuable rock from unwanted material
 - Extraction: of the metal rich rock from the ground
- The order in which these processes are carried out is:
- ☐ A concentration, extraction, refining.
 - ☐ B refining, extraction, concentration.
 - ☐ C concentration, refining, extraction.
 - ☐ D extraction, concentration, refining.
- 5** Aluminium is taken from its ore by a process involving the use of electricity. It is necessary to use such a process because:
- ☐ A aluminium is light, but does not burn.
 - ☐ B aluminium is not a very active metal.
 - ☐ C aluminium ore does not crumble easily.
 - ☐ D aluminium is a very active metal.
- 6** You are given a rock specimen and told that it is an ore. Because of its heavy weight, cubic-shaped crystals and silvery colour, you decide that it is:
- ☐ A coal.
 - ☐ B galena.
 - ☐ C bauxite.
 - ☐ D iron pyrite.
- 7** Which of the following is the best description of the metal lead?
- ☐ A Silvery grey colour, bends easily, fairly easily scratched, resists corrosion and very dense
 - ☐ B Silvery grey colour, very dense, resists corrosion and liquid at room temperature
 - ☐ C Silvery grey colour, bends easily, fairly easily scratched, resists corrosion and very low density
 - ☐ D Silvery grey colour, hard to bend, not easily scratched, corrodes readily and medium density
- 8** In Australia, commercial quantities of diamonds are found:
- ☐ A in the Blue Mountains.
 - ☐ B at Broken Hill.
 - ☐ C in the Kimberley Ranges.
 - ☐ D in the Darwin area.

- 9** An ore is a mineral that can be:
- ☐ A mined for profit.
 - ☐ B never used.
 - ☐ C used only in the manufacture of fuels.
 - ☐ D used only for space exploration.
- 10** The metals platinum and silver are mined because they are:
- ☐ A very good fuels.
 - ☐ B valuable metals.
 - ☐ C important sources of energy.
 - ☐ D used to refine crude oil.
- 11** One method of extracting a metal is to use coke and calcium carbonate and heat. Which of the following uses this process to extract the metal?
- ☐ A Extraction of iron from iron ore
 - ☐ B Extraction of aluminium from bauxite
 - ☐ C Purifying copper
 - ☐ D Making steel
- 12** Which of the following minerals is *not* mined at Broken Hill?
- ☐ A Silver
 - ☐ B Lead
 - ☐ C Rutile
 - ☐ D Zinc
- 13** Bauxite ores are formed in tropical climates. Which process forms these ores?
- ☐ A Volcanic eruptions
 - ☐ B Metamorphism
 - ☐ C Sedimentation
 - ☐ D Weathering
- 14** Iron ore from the Hammersley region in Western Australia was deposited by which process?
- ☐ A Weathering
 - ☐ B Sedimentation
 - ☐ C Glaciation
 - ☐ D Metamorphism
- 15** Coal is formed from organic material. Large deposits of coal are found in the Illawarra, Hunter, Bowen and Dalby districts. Several millions of years ago, these districts were:
- ☐ A flooded by seawater.
 - ☐ B swamps in tropical areas.
 - ☐ C woodlands on the edge of deserts.
 - ☐ D dry sclerophyll forests.
- 16** Uranium is a metal fuel that is used in:
- ☐ A hydro power stations.
 - ☐ B nuclear reactors.
 - ☐ C electricity sub-stations.
 - ☐ D motor vehicles
- 17** Malachite is a form of green copper ore and is known to react with acids, giving off carbon dioxide. Malachite is made from which compound?
- ☐ A Copper fluoride
 - ☐ B Copper oxide
 - ☐ C Copper carbonate
 - ☐ D Copper arsenate
- 18** Titanium is a corrosion-resistant metal that is mined from mineral sands. It reacts with oxygen to form a metal dioxide, which has many uses. Titanium dioxide is not used in:
- ☐ A sky writing.
 - ☐ B paint.
 - ☐ C sunscreen.
 - ☐ D cling wrap.
- 19** Mining is a way to extract ore from the ground. It includes open-cut, shaft mining and quarrying. Mining which of the following uses both open-cut and shaft mining?
- ☐ A Bauxite
 - ☐ B Coal
 - ☐ C Silver
 - ☐ D Lead

- 1** Storm surges are extremely hazardous and are caused by:
- A** a tsunami.
 - B** strong, driving winds and low atmospheric pressure in a cyclone.
 - C** a volcanic event.
 - D** high atmospheric pressure and a high tide after a cyclone.
- 2** Cyclones in tropical areas generate a lot of moisture which falls as torrential rain. Which of the following has a major impact after cyclonic rainfall?
- A** Flooding and landslides
 - B** Moderate turbidity of rivers and creeks
 - C** Leaf litter deposits
 - D** Deposition of sand in gutters
- 3** Which town in the Northern Territory did Cyclone Tracy destroy in 1974?
- A** Angurugu
 - B** Katherine
 - C** Nhulunbuy
 - D** Darwin
- 4** Every cyclone has an eye or centre where the wind is relatively calm in comparison to the outer part of the cyclone. Severe problems to settlements and the natural environment may be caused by:
- A** wind gusts that blow in different directions as the eye passes over the area.
 - B** wind gusts becoming less intense near the eye.
 - C** wind gusts becoming more intense near the eye.
 - D** wind gusts that blow in the same direction as the eye passes over the area.
- 5** Rare atmospheric events that have been known to occur in Australia during a cyclone are:
- A** Auroras.
 - B** ball and sheet lightning.
 - C** tornados.
 - D** sheet and forked lightning at different heights.
- 6** Excessive rainfall caused by cyclones can have a positive benefit. This is the:
- A** replenishment of aquifers.
 - B** washing away of rubbish left by humans.
 - C** removal of pollutants on the ground.
 - D** de-oxygenation of rivers.
- 7** Volcanoes release sulphur dioxide as aerosols. These aerosols can react with CFCs. The effect on the atmosphere is the:
- A** destruction of the ozone layer.
 - B** increase in the amount of hail formed.
 - C** decrease in cloud cover.
 - D** decrease in rainfall.
- 8** Volcanic ash during and after an eruption can alter climatic conditions on a global basis. This causes:
- A** an increase in rainfall patterns.
 - B** a decrease in global temperatures.
 - C** an initial decrease and then an increase in global temperatures.
 - D** a decrease in rainfall patterns that occurs only in the tropics.

9 Lava and volcanic ash assist agricultural production over many years by producing:

- A** rapid production of crops.
- B** acid soils.
- C** nutrients for plants.
- D** slow plant growth.

10 Tephra is a pyroclastic dust that is emitted from volcanoes. It is potentially dangerous because it:

- A** reacts with soil chemicals, killing micro-organisms.
- B** decreases the amount of animal fodder and contaminates water.
- C** heats the water and depletes micro-organisms.
- D** washes away soil that is needed by plants.

11 Some volcanic particles are water-soluble and hot. Which of the following factors can alter the biodiversity of aquatic life as a result of these particles?

- A** An increase in turbidity, temperature and alkalinity
- B** An increase in dissolved solids, temperature and acidity
- C** A decrease in turbidity, acidity and dissolved oxygen
- D** A decrease in temperature with an increase in turbidity and acidity

12 The Laki volcano eruption in 1783 had far-reaching global consequences. It caused starvation in Iceland and promoted social unrest in Europe and the United States. The main reason for the unrest was:

- A** an increase in winter temperatures that melted the snow.
- B** an increase in sunlight.
- C** a decrease in temperature that decreased agricultural production.
- D** a decrease in the length of daylight.

13 The tsunami in Indonesia in 2004 took many lives. The cause of this tsunami was:

- A** a cyclonic storm surge event.
- B** a volcanic eruption.
- C** an earthquake.
- D** a nuclear underwater test.

14 The city of Niigata in Japan was built on shallow, loose soils. When an earthquake struck this city, the soil was liquefied. Two major results of liquefaction are:

- A** displacement and tilting of buildings.
- B** making fissures and rock splitting.
- C** sand boiling and making the soil porous.
- D** lowering of ground altitude and making fissures.

15 Earthquakes may affect the hydrosphere by creating:

- A** highly acidic water flows.
- B** new lakes and displacing stream channels.
- C** sink holes in weathered granites.
- D** ionic disturbances in clouds.

16 Countries such as Pakistan and Iran have many earthquakes which result in major loss of life. This is due to:

- A** soil nutrient removal caused by the earthquake.
- B** severing power and gas lines.
- C** the change of the water supply by land movement.
- D** the density of the population and the brittle nature of housing.

17 Major earthquakes can affect the atmosphere when they strike in sedimentary basins, releasing gases. The gas that is most likely to be released is:

- ☐ A oxygen.
- ☐ B methane.
- ☐ C fluorine.
- ☐ D nitrogen trioxide.

18 Tens of thousands of people in Kashmir died due to an earthquake in 2005. A simple, cost-effective building method for mud brick homes to prevent earthquake damage would be the use of:

- ☐ A reinforced steel.
- ☐ B wire mesh and bamboo poles.
- ☐ C high quality concrete blocks.
- ☐ D galvanised iron.

19 Tiltmeters are used to record slope and distance changes when the ground swells. The swelling of the ground is due to which natural event?

- ☐ A Earthquakes
- ☐ B Volcanoes
- ☐ C Tsunamis
- ☐ D Atmospheric pressure

20 A monsoon trough is a low-pressure area and forms between two wind systems to form cyclones. In the Southern Hemisphere, cyclones rotate in which direction?

- ☐ A Anti-clockwise
- ☐ B From north to south
- ☐ C Clockwise
- ☐ D Vertically

21 Which of the following volcanoes is most likely to cause global cooling if it erupts again, erupting ash into the atmosphere?

- ☐ A Kilauea in Hawaii
- ☐ B Mt Pinatubo in the Philippines
- ☐ C Mount Berlin in Antarctica
- ☐ D Mt Kosciuszko in Australia

22 In 2005, Hurricane Katrina caused major flooding and damage to the levees that were in place in New Orleans in the United States. The main cause of the flooding was:

- ☐ A poorly engineered levees.
- ☐ B the fact that New Orleans is 1.5 metres below sea level in some areas.
- ☐ C the Mississippi River flooding because of the rain from Katrina.
- ☐ D the storm surge that came with the high tide as Katrina crossed the coast.

23 One positive aspect of bushfires in Australia is that they:

- ☐ A cause animals and plants to die off in large numbers.
- ☐ B assist in the regeneration of imported grasses.
- ☐ C assist in the regeneration of many different species of native trees.
- ☐ D contribute to the decline of flowering plants and Australian animals.

24 According to the Bureau of Meteorology, rainfall in some areas is increasing while in other areas it is decreasing. The reason for the increase in rainfall and storms is that:

- ☐ A there is more energy in the atmosphere due to climate change.
- ☐ B there is less energy in the atmosphere due to climate change.
- ☐ C there is more particulate matter in the atmosphere.
- ☐ D sea temperatures have changed.

- 1** All minerals can be mined for a profit. Which of the following is not profitable?
- A** Small quantities of coal for power stations
 - B** Small quantities of gold for jewellery
 - C** Small quantities of tellurium for mobile phones
 - D** Large volumes of clay for bricks
- 2** Sustainable development refers to development that:
- A** only occurs once.
 - B** leaves no resources.
 - C** leaves resources for future use.
 - D** is based on oil products.
- 3** Water conservation is essential, particularly for potable water. Globally, this is necessary because:
- A** there is a high demand for water in Australia.
 - B** there is a drought in Australia.
 - C** water evaporates and is lost to space.
 - D** the supply of drinkable, fresh water in the world is extremely limited.
- 4** Approximately 45% of typical household garbage is food scraps. It is best to:
- A** have them removed and dumped in a council tip.
 - B** compost them to make better soil.
 - C** have them removed and burnt.
 - D** have them flushed out to sea.
- 5** Coal and oil are often described as fossil fuels as they:
- A** have been made from ancient liquids.
 - B** are made from compressed inorganic matter.
 - C** were made from compressed organic substances millions of years old.
 - D** were formed from compressed organic substances thousands of years old.
- 6** The first metals made and used by humans 3000 years ago were:
- A** gold, silver and copper.
 - B** lead, copper and aluminium.
 - C** tin, aluminium and palladium.
 - D** silver, lead and zinc.
- 7** Substances such as haematite and bauxite are mined because they are:
- A** very good fuels.
 - B** useful metal ores.
 - C** important sources of energy.
 - D** used to refine crude oil.
- 8** The fuels used by people for cooking include wood, oil, coal and gas. What do these fuels all have in common?
- A** They will run out by the year 2040.
 - B** They are renewable resources.
 - C** They are fuels containing little energy.
 - D** They release carbon dioxide when they are burnt.

- 9** One method of extracting a particular metal is to use a blowpipe to heat lead oxide with charcoal, forming a ball of lead. This is similar to the way in which:

A iron is separated from iron ore.
B aluminium is obtained from bauxite.
C impure copper is made pure.
D iron is converted into steel.

- 10** Which of the following materials obtained from the Earth is not used in buildings?

A Limestone
B Sandstone
C Clay
D Coal

- 11** The cost of petrol increases annually because:

A the unions go on strike.
B there is a blockade in the Middle East.
C demand matches supply.
D it is a limited resource.

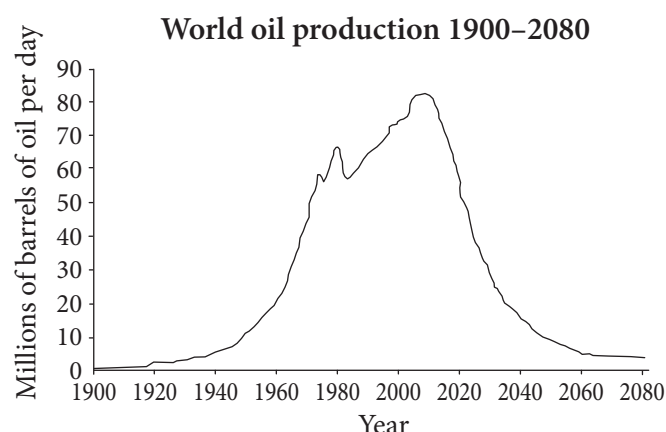
- 12** A number of processes are used to extract and refine oil. Four of these processes are listed below.

- Fractional distillation – separating the oil into its parts.
- Drilling – making an oil well through which oil is pumped from the ground.
- Additives – mixing special chemicals with the oil or petrol before it is sold to the public.
- Cracking – converting less useful parts into more useful products.

The order in which these processes take place is:

A a, b, c, d
B b, a, d, c
C b, d, a, c
D d, c, b, a

Questions 13 and 14 are based on the information provided in the graph.



Source: Australian Association for the Study of Peak Oil and Gas

- 13** How many barrels of oil will be produced each day in the year 2020?

A 50 million. **B** 75 million.
C 60 million. **D** 80 million.

- 14** According to the graph, the year in which oil production will start to decline will be:

A 1999. **B** 2050.
C 2000. **D** 2010.

- 15** Production of electricity by hydro-electric generation is cheaper than burning oil because:

A water is scarce.
B hydro-electric generation is as costly as oil-fired power stations.
C water is not a fossil fuel.
D water can drive generators at less cost.

- 16** Electricity made by wind energy is:

A very economical and clean.
B is useful where there is variable wind.
C is still in its experimental phase.
D is economic only when supported by solar energy.

- 1** Bauxite is aluminium ore that has been refined electrically using carbon anodes. The refining material is sodium aluminium fluoride and steam combined with aluminium oxide. The waste gases produced by this electrical refining process are:
- ☐ A carbon dioxide and fluorine gas.
 - ☐ B sodium fluoride and carbon monoxide.
 - ☐ C water vapour and carbon.
 - ☐ D carbon dioxide and soot.
- 2** In tertiary sewage treatment plants, the solid waste is treated to remove bacteria, viruses and other harmful products. The solid waste is then used as:
- ☐ A road fill.
 - ☐ B plant potting mix.
 - ☐ C cement filler.
 - ☐ D embedding material for sewer pipes.
- 3** Paper-making requires many megalitres of water. Much of this water is reused and recycled. The water is collected from which paper-making by-product?
- ☐ A Dry cellulose fibre
 - ☐ B Sludge from the pulping process
 - ☐ C Black dyes in printing
 - ☐ D Shredded lignin
- 4** Blast furnace slag composition includes calcium oxide, magnesium oxide, sulphur, silicon dioxide, manganese oxide and aluminium oxide. Which of these by-products is in its elemental molecular form?
- ☐ A Aluminium oxide
 - ☐ B Manganese oxide
 - ☐ C Silicon dioxide
 - ☐ D Sulphur
- 5** After grinding, blast furnace slag is used in which product?
- ☐ A Soap
 - ☐ B Cement
 - ☐ C Polishing wax
 - ☐ D Clay
- 6** Plastics are made from carbon and hydrogen. They are often pelletised and used in blast furnaces as a reducing agent. The gases produced in the blast furnace are carbon monoxide and methane, which are collected for use in other facilities. Which facility can use methane gas for combustion?
- ☐ A Sewage treatment plants
 - ☐ B Power stations
 - ☐ C Waste product recycling plants
 - ☐ D Landfill operations
- 7** Nuclear reactors produce many isotopes that are used in medicine, the construction industry, armaments and power plants. These radioactive materials can last from several seconds to millions of years. Which of the following has the longest radioactive half-life?
- ☐ A Caesium-137
 - ☐ B Uranium-238
 - ☐ C Tin-126
 - ☐ D Iodine-129
- 8** There are 78 radioactive substances that have polluted the environment since the atomic bomb was dropped in 1945. Other bombs have been dropped off the Australian coast, Maralinga, the French Polynesian islands and Bikini Atoll. The resulting radioactive substances are found in which parts of the environment?
- ☐ A Ground water only
 - ☐ B Ground water, air and soil
 - ☐ C Air only
 - ☐ D Soil only

- 9** In Australia, many batteries are thrown into the household rubbish and then transported to a landfill site. This is a problem because the batteries produce:
- A** a non-toxic powder that will react with other rubbish.
 - B** a toxic leachate that flows into the environment.
 - C** a harmless liquid that flows into the environment.
 - D** a chemical that produces methane gas.

- 10** According to the Federal Government, in 2006 86% of the population in Australia recycled many materials. However, the government was concerned that while facilities existed for recycling, much of the material put out for recycling was sent to landfill, as it was economically cheaper. The table below indicates household rubbish that was not recycled.

Kitchen or food waste	20.6
Old clothing	12.4
Garden waste	12.4
Rags	2.1
Glass	15.7
Paper	17.5

Source: Australian Bureau of Statistics *Environmental Issues/Peoples Views and Practices* (4602.0) for 2006

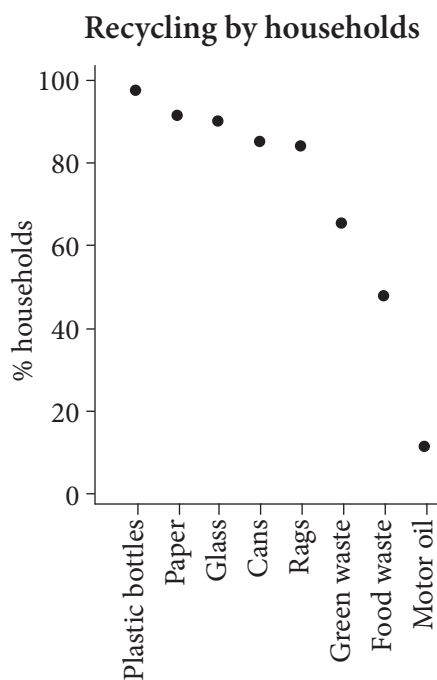
The main reason this rubbish was not recycled is:

- A** insufficient services.
 - B** no storage facility in the household.
 - C** insufficient infrastructure for recycling.
 - D** too difficult to compost.
- 11** Materials from various sources can be converted into electricity. Which of the following can be used to make electricity?
- A** car emissions
 - B** recycled biological waste
 - C** radioactive waste
 - D** soil

- 12** It is estimated that 91% of oil is disposed of properly. Transfer stations collect 4% and collection points and household rubbish collect 4%. Small amounts of oil are sometimes poured down the drain. Which percentage represents the oil that is poured down the drain?

- A** 4%
- B** 8%
- C** 1%
- D** 6%

- 13** The graph shows materials that are recycled by households.



Source: Australian Bureau of Statistics *Environmental Issues/Peoples Views and Practices* (4602.0) for 2006

Which of the following materials are most recycled?

- A** Plastics, oil, rags, glass
- B** Paper, green waste, glass, rags
- C** Oil, green waste, paper, cans
- D** Plastics, paper, glass, cans

14 Waste from households goes to landfills, where the rubbish is processed and buried to reduce the amount of carbon dioxide produced and allow bacteria to digest the buried products. Which of the following provides a reduction in the levels of the greenhouse gas carbon dioxide in the landfill?

- ☐ A Aerobic bacterial digestion
- ☐ B Algae pond digestion
- ☐ C Anaerobic bacterial digestion
- ☐ D Methane production from bio-digestion

15 Next to water, soil is one of Australia's most precious resources. However, some farming practices from the days of settlement to the present time reduce the amount of soil available for agriculture. This is due to which management practice?

- ☐ A Deforestation
- ☐ B Reafforestation
- ☐ C Draining of wetlands
- ☐ D Under-stocking of livestock

16 Prior to the 1960s, mining in Australia was the cause of several environmental disasters. Now environmental assessment is done before mining begins. The waste and topsoil are stockpiled. Stockpiling of topsoil is a requirement and is useful for:

- ☐ A establishing new mines.
- ☐ B preventing the loss of minable material.
- ☐ C rehabilitating the land after mining ceases.
- ☐ D providing material for roads.

17 Many of Australia's food crops are produced by irrigation, but this process causes a loss in soil fertility which is due to:

- ☐ A over cropping.
- ☐ B insufficient fertiliser usage.
- ☐ C dissolved salts rising to the surface.
- ☐ D excessive water use.

18 Several metal sulphide ores have other elements in them that can be detrimental to organisms and the environment. Many mining companies use flotation, oxidation, carbonation and metal reactivity to separate the different metals. The advantage of this new technology is that it reduces the amount of which gas from entering the atmosphere?

- ☐ A Tetra fluoro decane gas
- ☐ B Carbon dioxide gas
- ☐ C Hydrogen fluoride gas
- ☐ D Nitrogen gas

19 Two of the residual elements in lead sulphide ore are extracted and used in the jewellery industry. These two elements are:

- ☐ A silver and gold.
- ☐ B gold and antimony.
- ☐ C copper and bismuth.
- ☐ D arsenic and silicon dioxide.

20 Crude oil wastes are broken down into other compounds by a process called catalytic cracking. Two residual products that are made from cracking and are widely used are:

- ☐ A plastics and cyclohexane.
- ☐ B waxes and bitumen.
- ☐ C hydrogen sulphide gas and polystyrene.
- ☐ D naphthalene and xylenes.

Topic 1 – The solar system

1 B 2 B 3 D 4 D 5 A
6 B 7 D 8 B 9 D 10 C
11 D 12 C 13 C 14 B 15 B
16 A 17 D 18 A 19 A 20 A

Topic 2 – Objects in the galaxy

1 D 2 C 3 C 4 D 5 A
6 B 7 D 8 C 9 A 10 A
11 B 12 C 13 A 14 A 15 C
16 A 17 B 18 B 19 D 20 C
21 D

Topic 3 – Space programs

1 B 2 D 3 C 4 D 5 A
6 D 7 C 8 A 9 C 10 D
11 B 12 B 13 A 14 C 15 B
16 B 17 C 18 B 19 A 20 B
21 A

Topic 4 – Origin of the universe and electromagnetic radiation

1 C 2 A 3 C 4 A 5 D
6 D 7 B 8 A 9 B 10 C
11 A 12 A 13 B 14 C 15 D
16 C 17 A 18 D 19 D 20 A
21 A 22 C 23 B

Topic 5 – Obtaining information about the universe

1 B 2 A 3 C 4 D 5 D
6 D 7 B 8 A 9 A 10 D
11 D 12 C 13 C 14 C 15 D
16 A 17 A 18 D 19 B 20 B

Topic 6 – Major features of the universe and life of a star

1 A 2 B 3 A 4 A 5 C
6 C 7 A 8 A 9 B 10 C
11 D 12 A 13 B 14 A 15 A
16 D 17 D 18 A 19 A

Topic 7 – Earth movements

1 A 2 D 3 B 4 A 5 B
6 B 7 D 8 A 9 A 10 C
11 C 12 B 13 B 14 D 15 A
16 C 17 C 18 C 19 C 20 D
21 D 22 B

Topic 8 – Fossils and extinction

1 A 2 A 3 D 4 B 5 C
6 C 7 C 8 A 9 B 10 B
11 C 12 B 13 B 14 A 15 B
16 C 17 B 18 B 19 C 20 C
21 D 22 C

Topic 9 – Fossil fuels and the greenhouse effect

1 B 2 A 3 C 4 B 5 B
6 D 7 B 8 B 9 C 10 D
11 C 12 C 13 C 14 C 15 D
16 C 17 B 18 C 19 C 20 B
21 C 22 A

Topic 10 – Geological time

1 B 2 A 3 C 4 C 5 D
6 C 7 D 8 B 9 B 10 C
11 C 12 B 13 D 14 A 15 A
16 B 17 B 18 D 19 A 20 C

Topic 11 – Human impact

1 B 2 D 3 B 4 C 5 B
6 B 7 C 8 B 9 D 10 C
11 B 12 B 13 B 14 B 15 C
16 D 17 B 18 B

Topic 12 – Minerals

1 C 2 C 3 A 4 D 5 D
6 B 7 A 8 C 9 A 10 B
11 A 12 C 13 D 14 B 15 B
16 B 17 C 18 D 19 B

Topic 13 – Impacts of natural events

1 B 2 A 3 D 4 A 5 C
6 A 7 A 8 B 9 C 10 B
11 B 12 C 13 C 14 A 15 B
16 D 17 B 18 B 19 B 20 C
21 B 22 D 23 C 24 A

Topic 14 – Resources and sustainability

1 A 2 C 3 D 4 B 5 C
6 A 7 B 8 D 9 A 10 D
11 D 12 B 13 C 14 D 15 D
16 A

Topic 15 – Waste from resource use

1 A 2 B 3 B 4 D 5 B
6 B 7 B 8 B 9 B 10 C
11 B 12 C 13 D 14 C 15 A
16 C 17 C 18 B 19 A 20 B

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